

# Chapter 7 Latches, Flip-Flops, and Timers

This is

# THE END

of Combinatorial Circuits

and

# the beginning

of sequential circuits

# Sequential Circuits

- The output of a **Combinatorial Circuit** depends only on the current inputs
- The output of a **Sequential Circuit** depends on the current and past inputs

# Outline

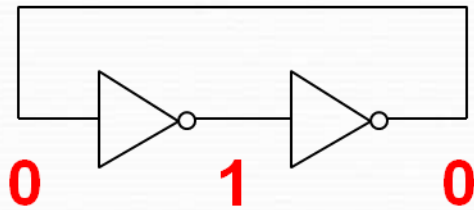
- Latches （锁存器）
- Edge-triggered Flip-Flops （触发器）
- Flip-Flop Operating Characteristics
- Flip-Flop applications
- One-shots （单稳态触发器）
- The 555 timer （多谐振荡器）

# 7.1 Latches

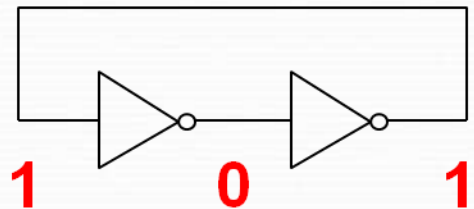
- A type of temporary storage device
- Have two stable states

# Bi-Stability

## The Key to Memory

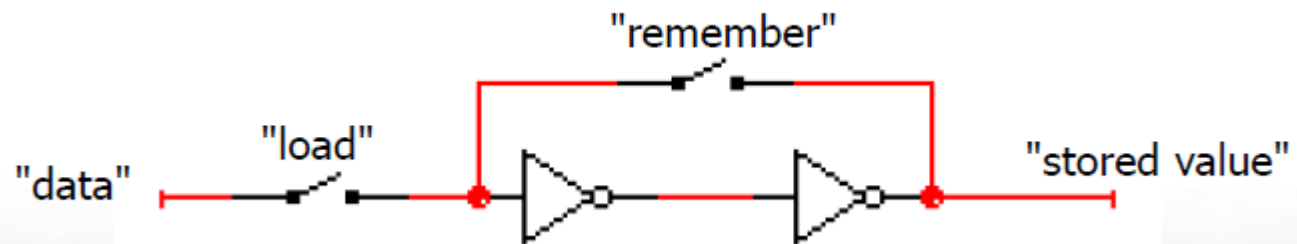


This is a stable state -  
it will sit like this forever

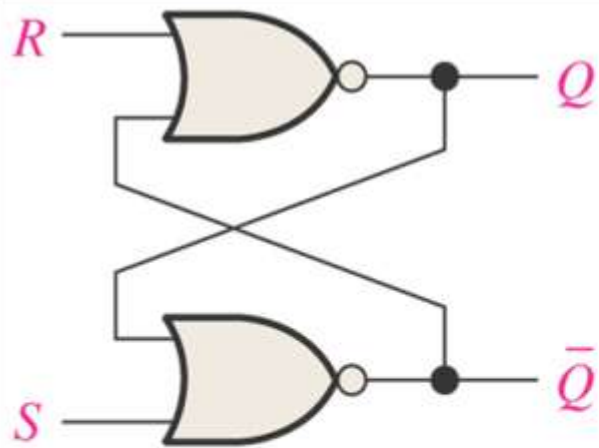


This is also a stable state -  
it will sit like this forever

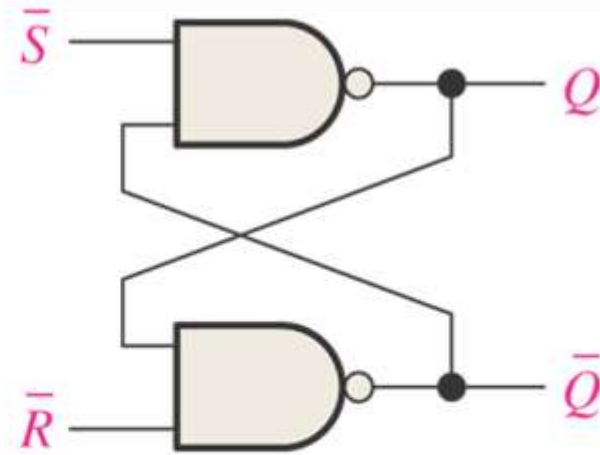
There are 2 stable states -  
a **bi-stable** circuit...



# The S-R (Set-Reset) Latch



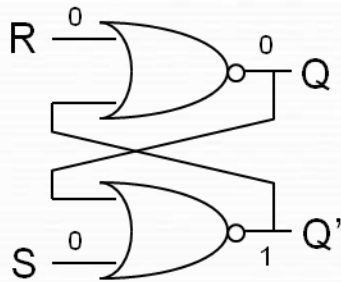
(a) Active-HIGH input S-R latch



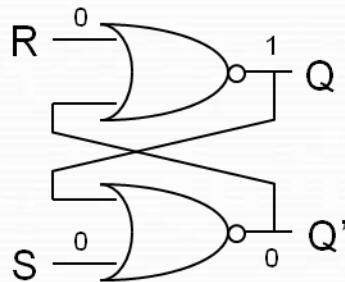
(b) Active-LOW input  $\bar{S}$ - $\bar{R}$  latch

In normal operation, the outputs of a latch are always complements of each other.

# SR Latch - A Bi-Stable Circuit

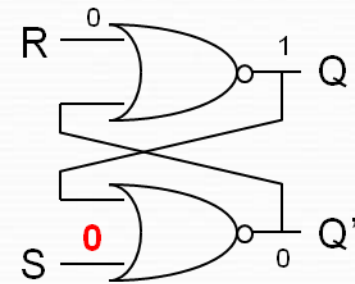
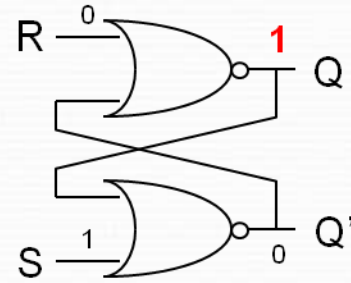
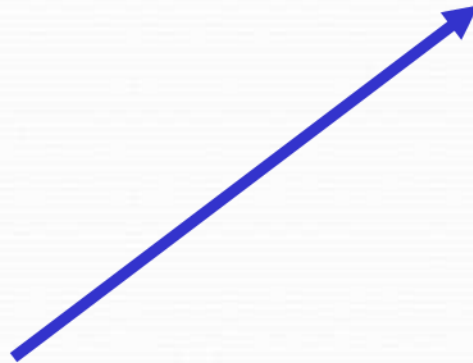
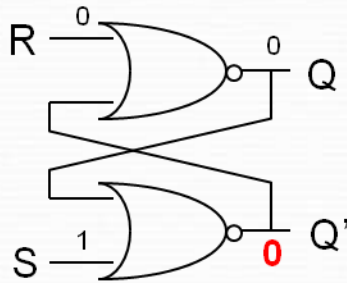
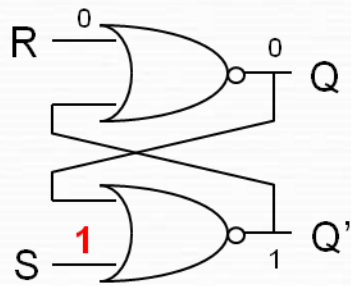


This is a stable state -  
it will sit like this forever



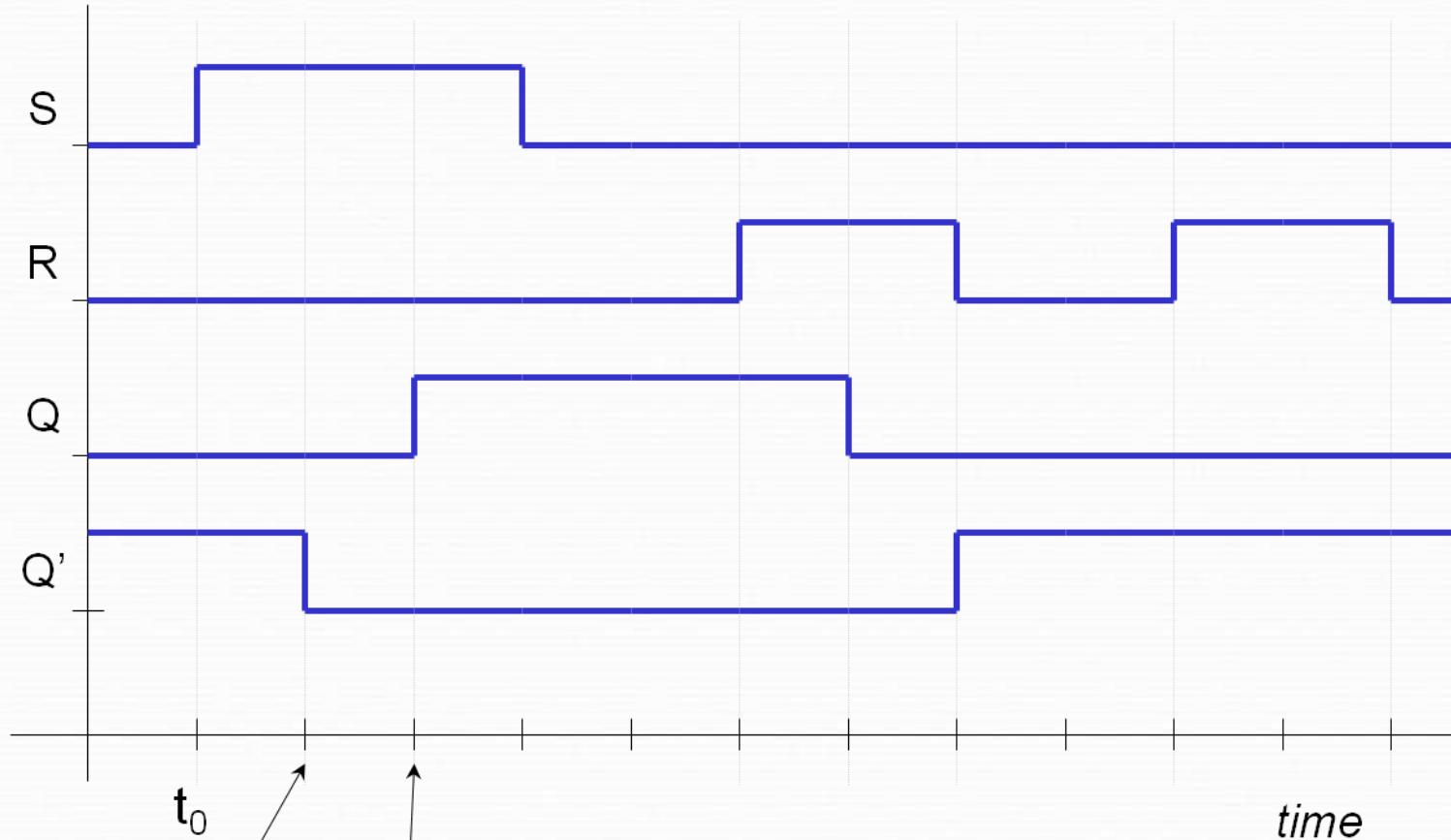
This is also a stable state -  
it will sit like this forever

# SR Latch Transition

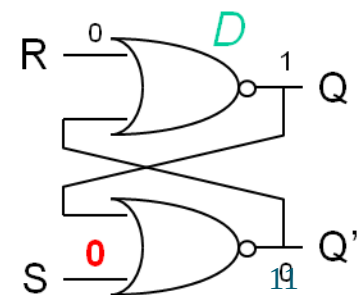
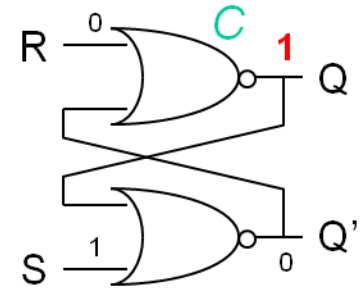
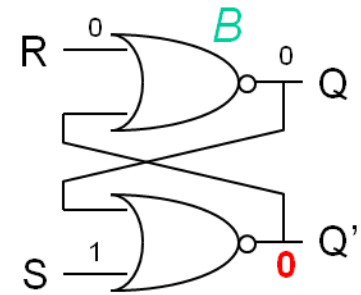
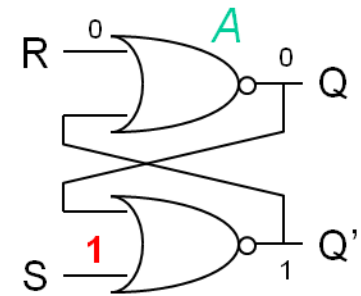


# SR Latch Timing

A B C D

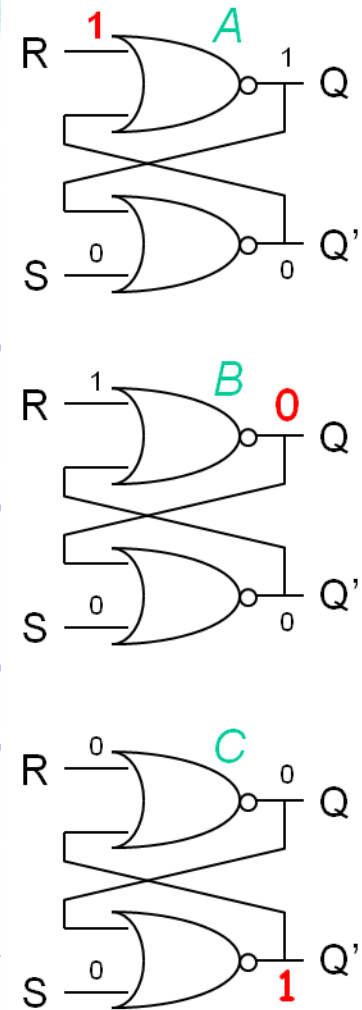
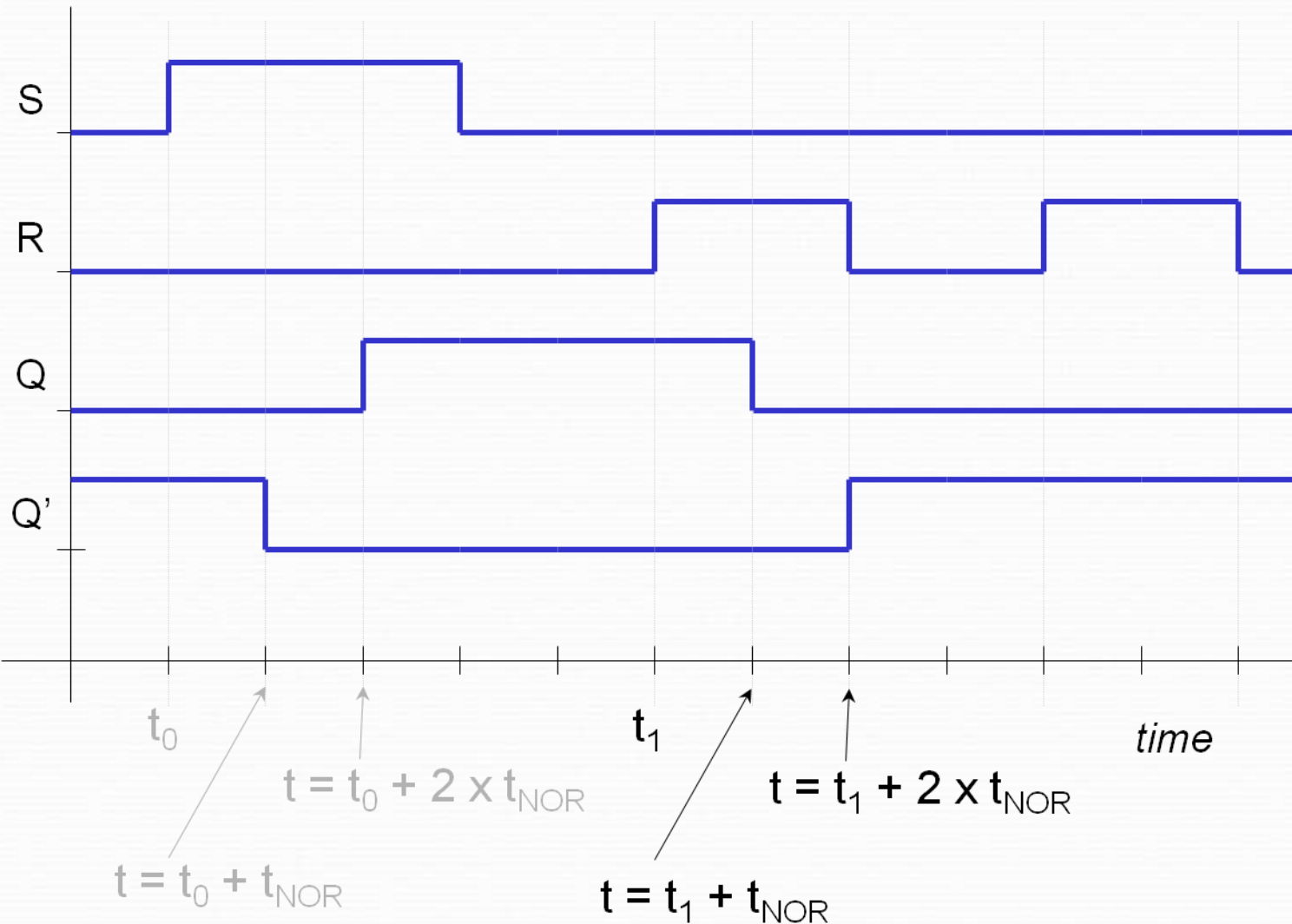


$t_0$   
 $t = t_0 + t_{\text{NOR}}$   
 $t = t_0 + 2 \times t_{\text{NOR}}$

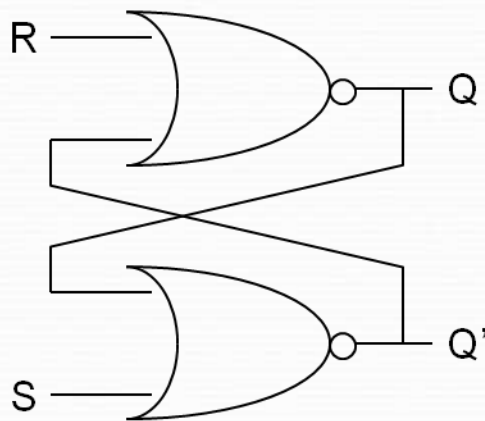


# SR Latch Timing

A B C



# SR Latch Transition Table

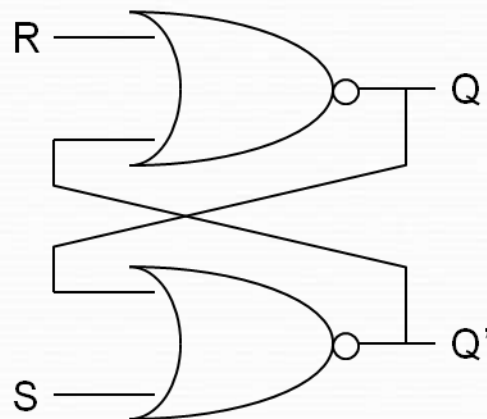


| S | R | Q | Q+ |
|---|---|---|----|
| 0 | 0 | 0 | 0  |
| 0 | 0 | 1 | 1  |
| 0 | 1 | 0 | 0  |
| 0 | 1 | 1 | 0  |
| 1 | 0 | 0 | 1  |
| 1 | 0 | 1 | 1  |
| 1 | 1 | 0 | ?  |
| 1 | 1 | 1 | ?  |

The current state of the Q output

The next state of the Q output  
(what it will change to)...

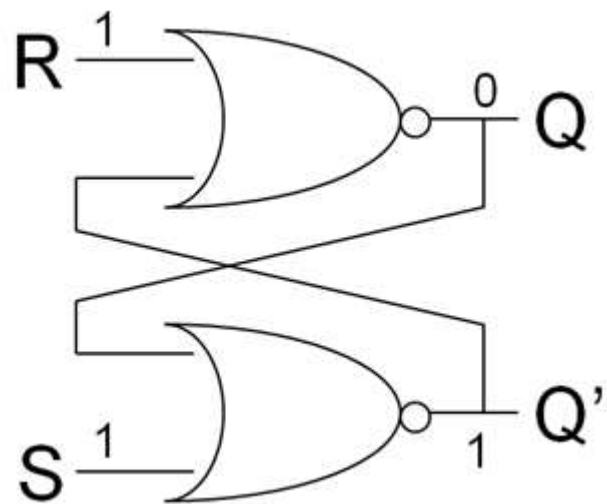
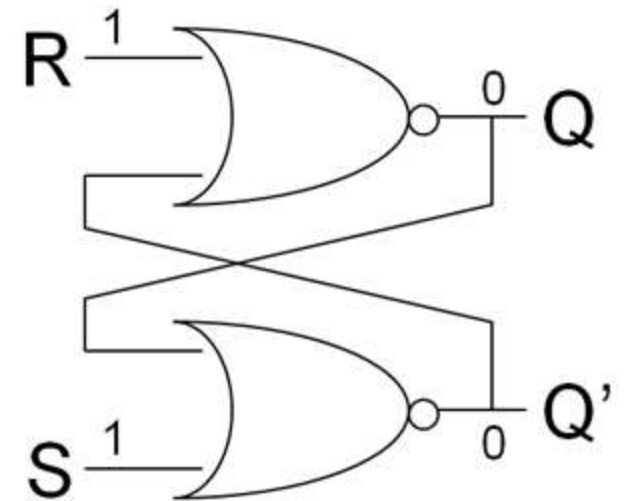
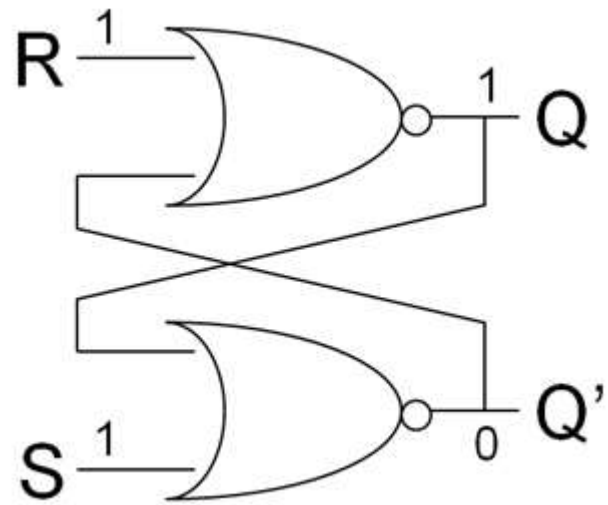
# SR Latch Transition Table

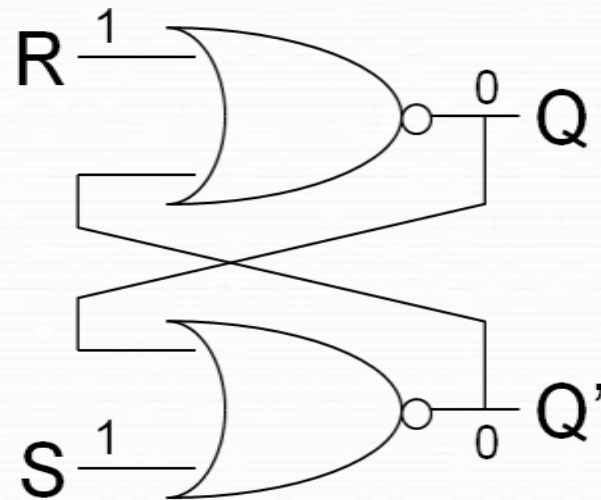


| S | R | Q | Q+ |           |
|---|---|---|----|-----------|
| 0 | 0 | 0 | 0  | No change |
| 0 | 0 | 1 | 1  |           |
| 0 | 1 | 0 | 0  | Reset it  |
| 0 | 1 | 1 | 0  |           |
| 1 | 0 | 0 | 1  | Set it    |
| 1 | 0 | 1 | 1  |           |
| 1 | 1 | 0 | ?  |           |
| 1 | 1 | 1 | ?  |           |

The current state of the Q output

The next state of the Q output  
(what it will change to)...

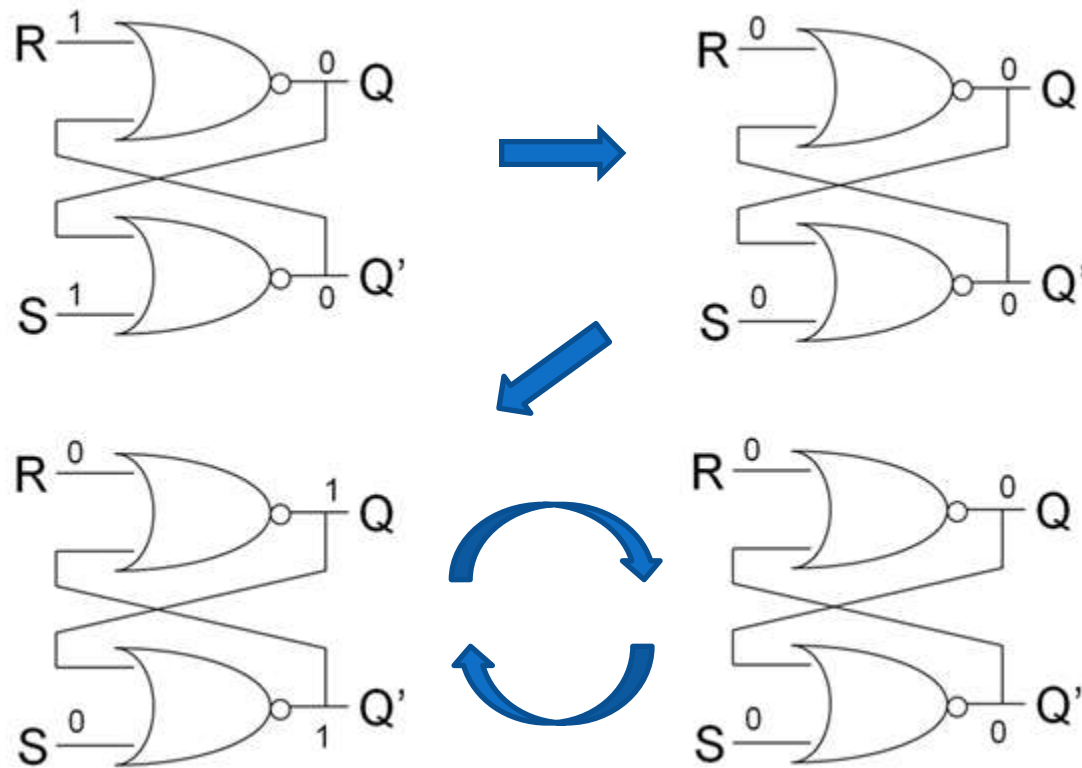




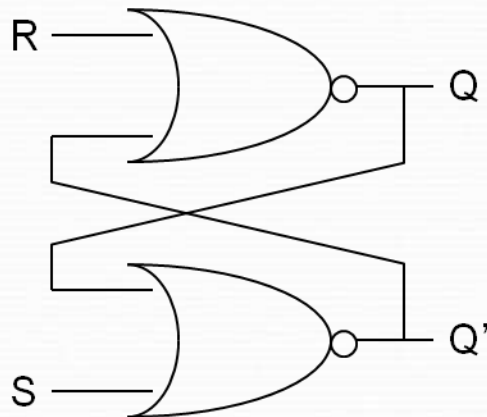
Is the latch *SET*??? → no  
Is the latch *RESET*??? → no  
What is it? → neither

We avoid this input combination in normal usage,  
mainly because it makes no sense.

# Assuming perfect synchronization



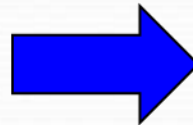
# SR Latch Transition Table



| S | R | Q | Q+  |           |
|---|---|---|-----|-----------|
| 0 | 0 | 0 | 0   | No change |
| 0 | 0 | 1 | 1   |           |
| 0 | 1 | 0 | 0   | Reset it  |
| 0 | 1 | 1 | 0   |           |
| 1 | 0 | 0 | 1   | Set it    |
| 1 | 0 | 1 | 1   |           |
| 1 | 1 | 0 | N/A | Unused    |
| 1 | 1 | 1 | N/A |           |

# SR Latch - Next State Equation

| S | R | Q | Q+ |
|---|---|---|----|
| 0 | 0 | 0 | 0  |
| 0 | 0 | 1 | 1  |
| 0 | 1 | 0 | 0  |
| 0 | 1 | 1 | 0  |
| 1 | 0 | 0 | 1  |
| 1 | 0 | 1 | 1  |
| 1 | 1 | 0 | X  |
| 1 | 1 | 1 | X  |



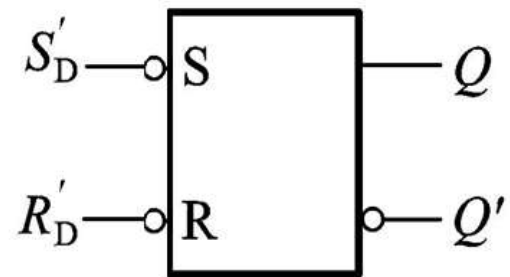
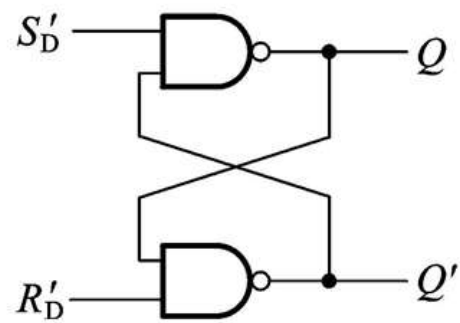
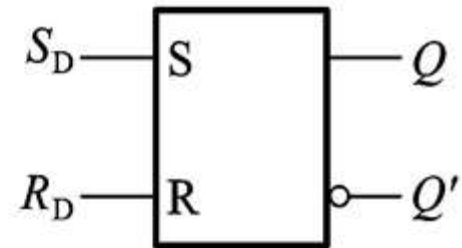
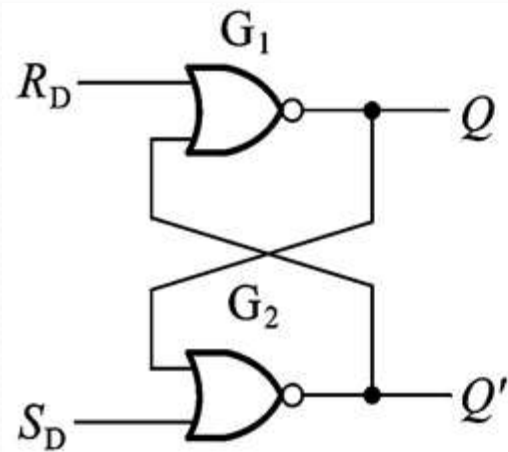
|   |   | SR |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| Q | 0 |    |    | X  | 1  |
|   | 1 | 1  |    | X  | 1  |

$$Q+ = S + \bar{R}Q$$

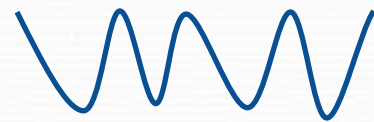
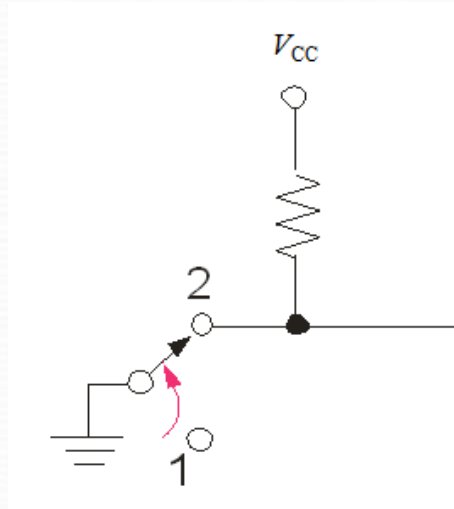
- The next state of Q will be true any time:
- You SET it
  - It is already true and you don't RESET it

# SR Latches

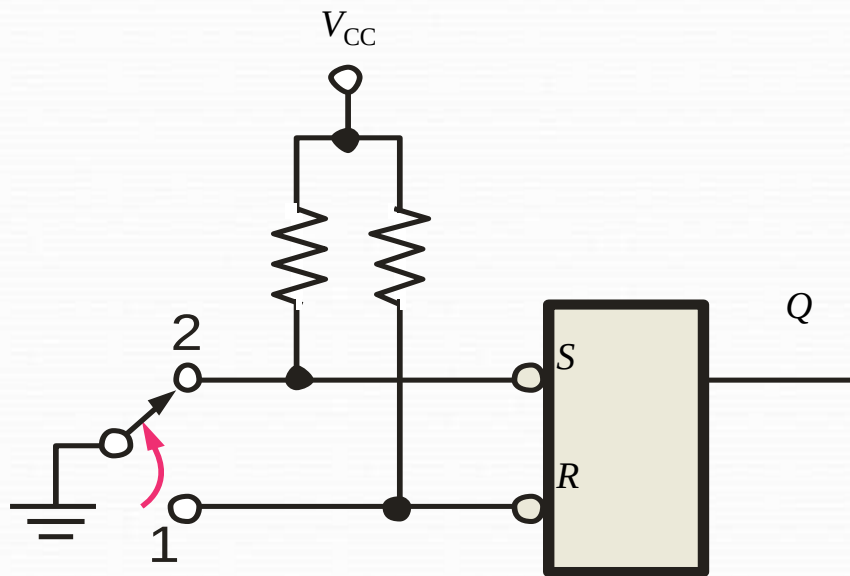
- Illustrate simple notion of *bi-stability*
  - Two stable states
  - S and R inputs move latch between them
- A memory
  - When  $Q=1 \Leftrightarrow$  latch storing a '1'
  - When  $Q=0 \Leftrightarrow$  latch storing a '0'
- Will hold its value indefinitely
  - As long as circuit is powered



# An Application of S-R Latch



Fluctuating voltage  
due to contact bounce

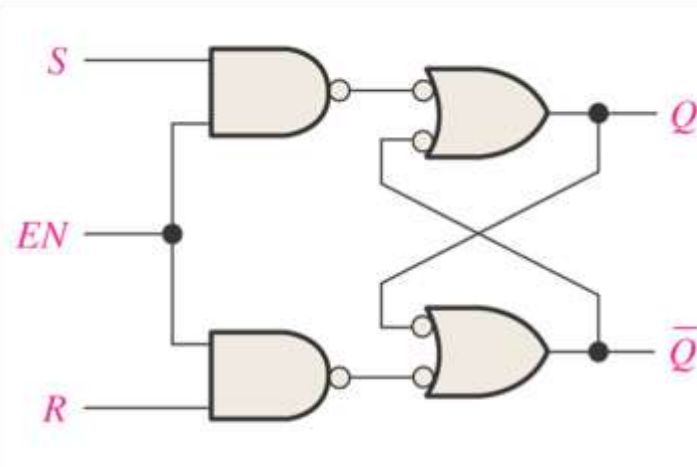


# Gated Latches

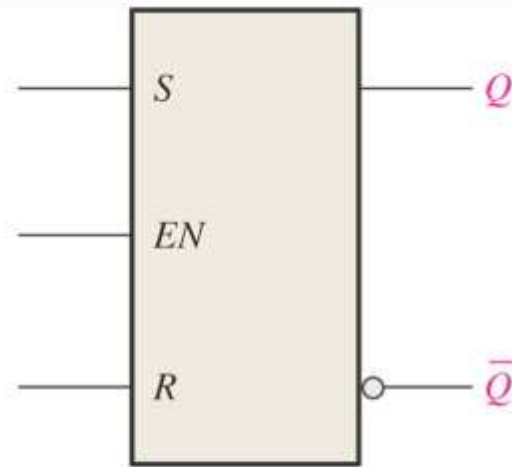
# SR Latches Always Sampling Inputs

- An SR latch will always respond to S/R input changes
  - ALWAYS!
- Sometimes we want to control when a storage element changes
  - Use a *gated latch*

# The Gated SR Latch



(a) Logic diagram

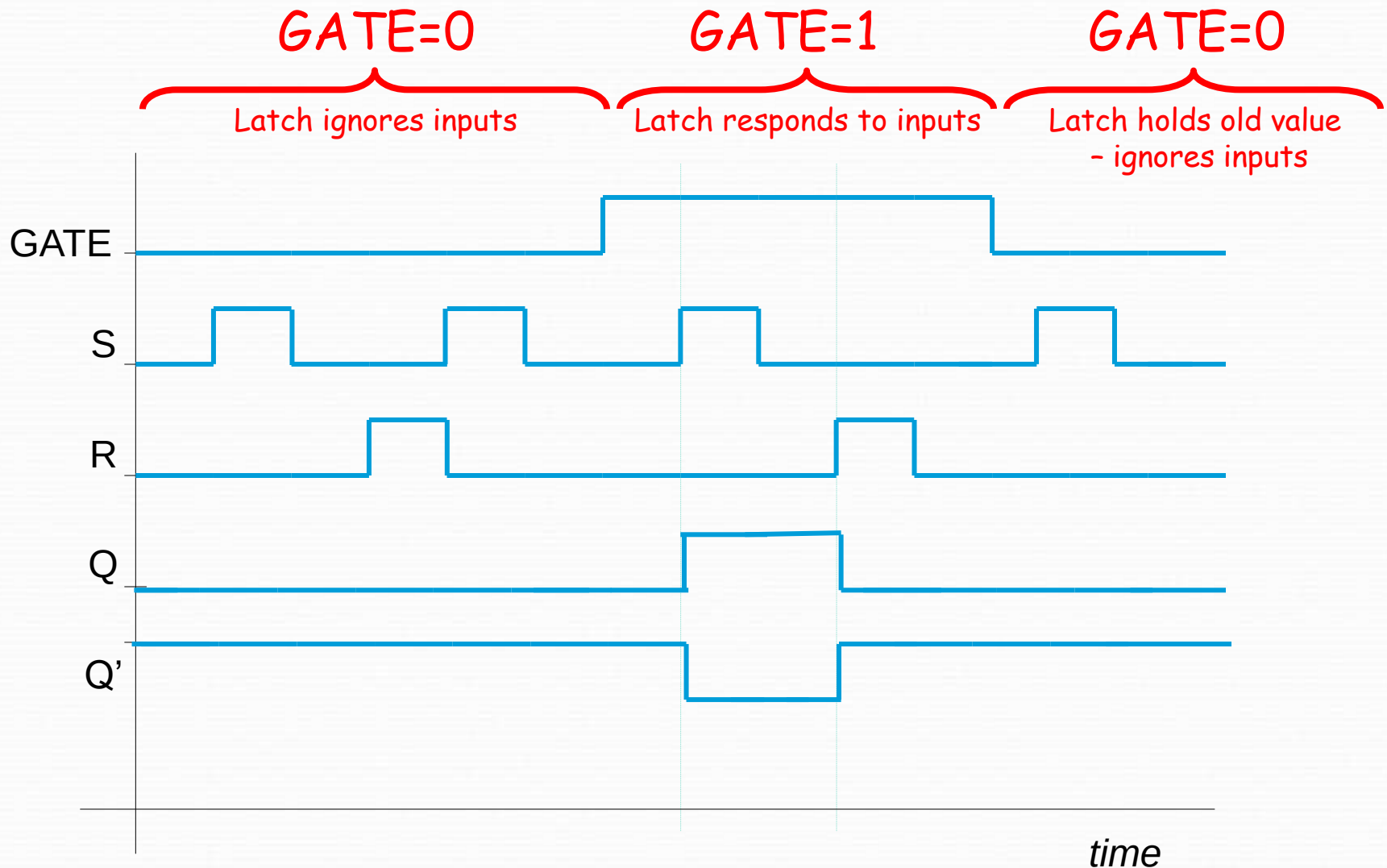


(b) Logic symbol

- When  $EN = 0$ 
  - Latch cannot be modified
- When  $EN = 1$ 
  - Works like an SR latch

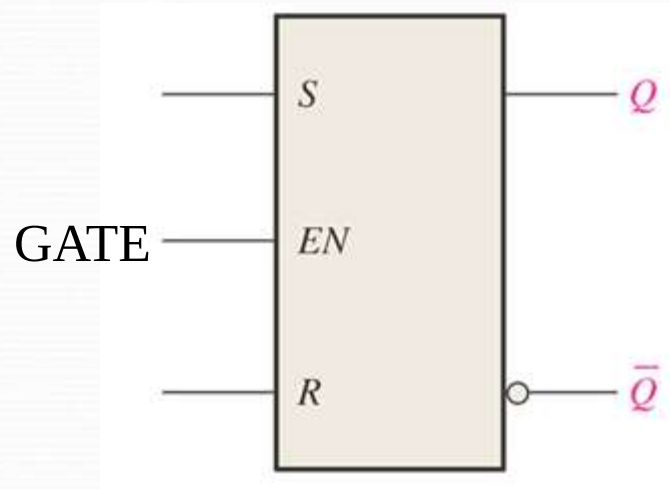
The  $EN$  signal allows us to control *when* the latch will be loaded with a new value

# Gated SR Latch Timing

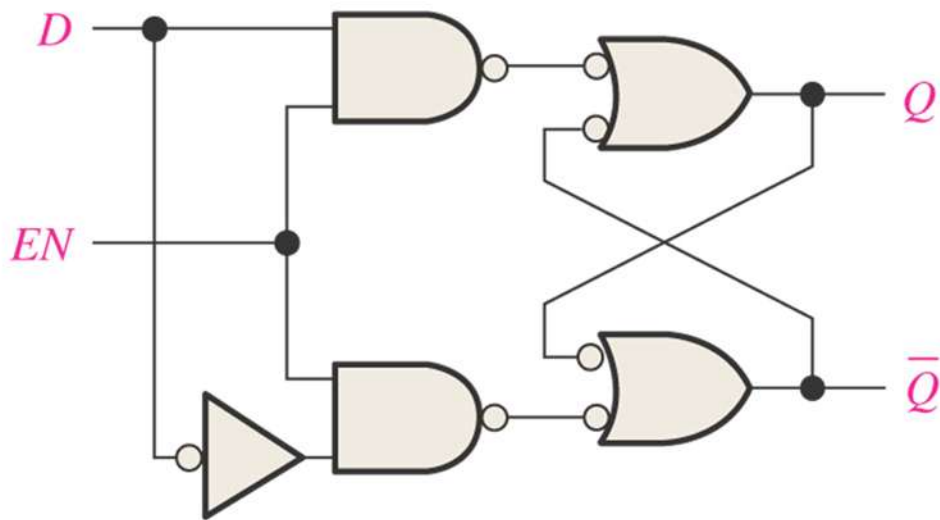


# Gated SR Latch

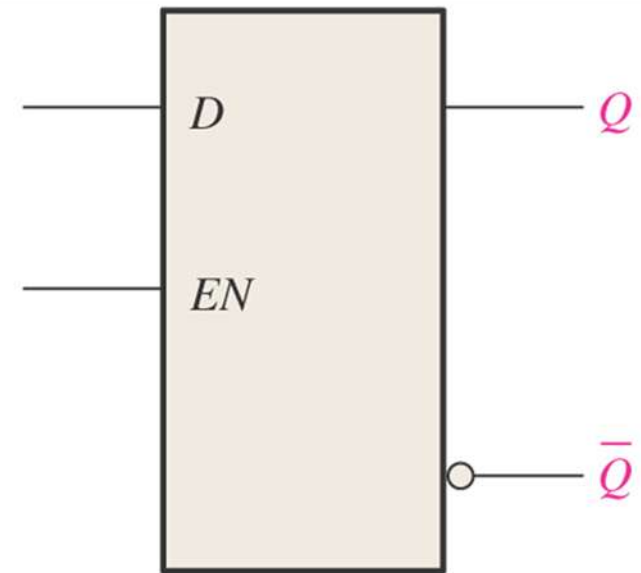
- Sometimes known as a *loadable SR latch*
  - The GATE signal acts like a *load* input



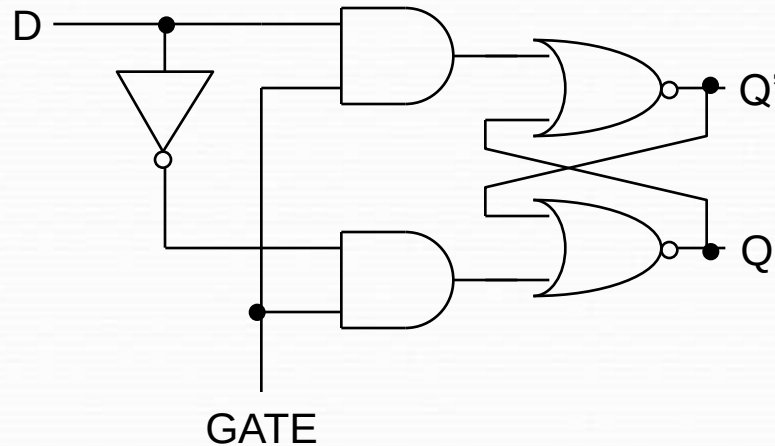
# The Gated D Latch



(a) Logic diagram



(b) Logic symbol

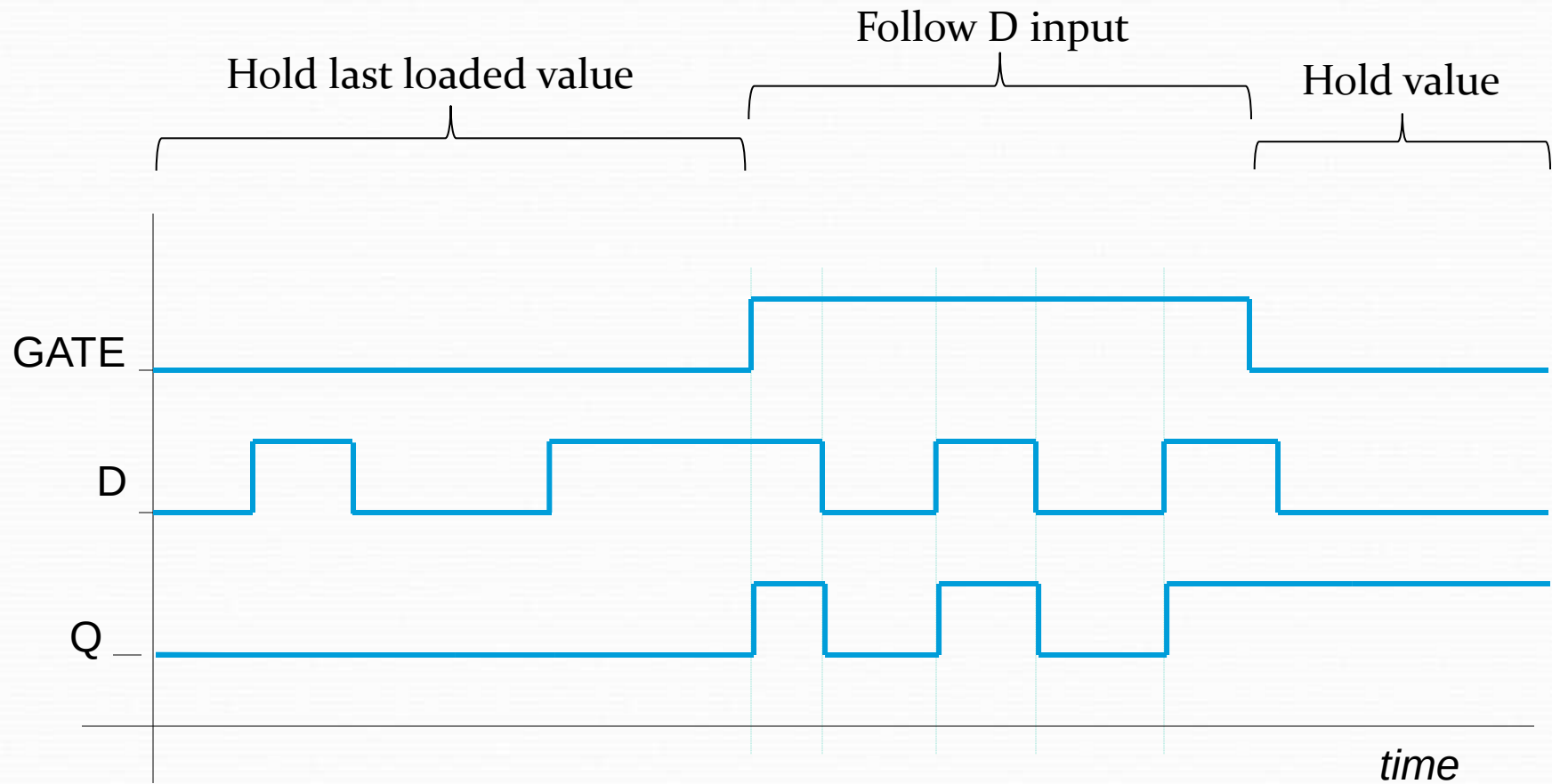


$$Q^{n+1} = GATE \cdot D + \overline{GATE} \cdot Q^n$$

| GATE | D | Q | $Q^{n+1}$ |
|------|---|---|-----------|
| 0    | 0 | 0 | 0         |
| 0    | 0 | 1 | 1         |
| 0    | 1 | 0 | 0         |
| 0    | 1 | 1 | 1         |
| 1    | 0 | 0 | 0         |
| 1    | 0 | 1 | 0         |
| 1    | 1 | 0 | 1         |
| 1    | 1 | 1 | 1         |

When  $GATE = 1 \Leftrightarrow Q$  follows  $D$  (storage)  
 When  $GATE = 0 \Leftrightarrow Q$  retains old value (retention)

# Gated D Latch - Timing



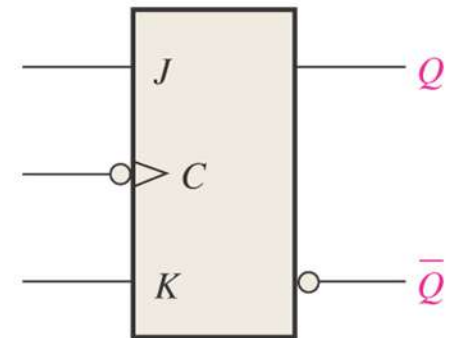
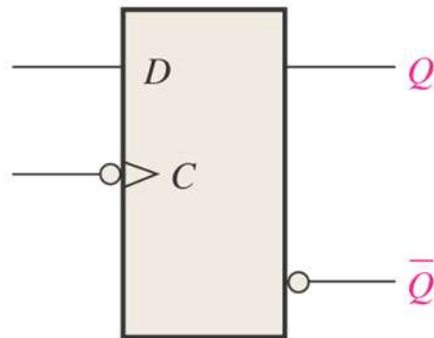
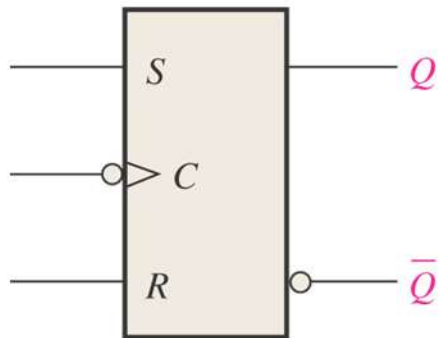
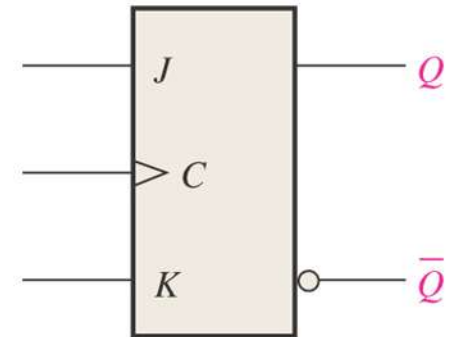
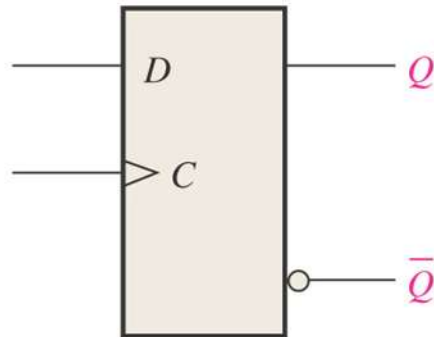
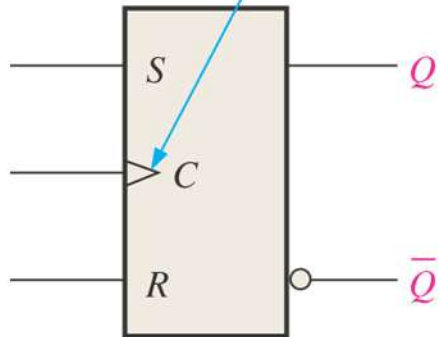
# Gated D Latches

- Sometimes called a *transparent* latch
  - When  $GATE = 1$ , the D input propagates to Q output
- Allows us to control *when* to store new data into latch
  - D = data to be stored
  - GATE = control signal

## 7.2 Edge-Triggered Flip-Flops

- Edge-triggered flip-flops
  - Change state either at the positive edge or at the negative edge of the clock
  - Sensitive to inputs only at the transition of the clock
- Three types
  - S-R, D, and J-K

Dynamic input  
indicator

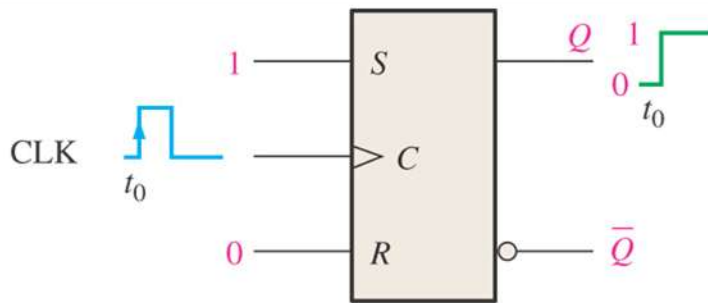


(a) S-R

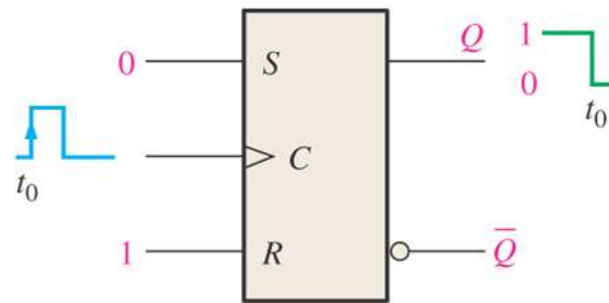
(b) D

(c) J-K

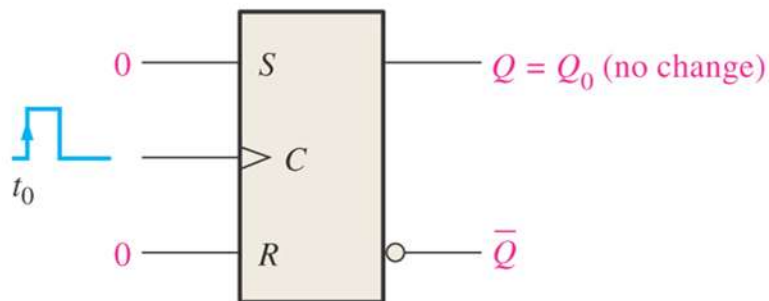
**Figure 7-14** Operation of a positive edge-triggered S-R flip-flop.



(a)  $S = 1, R = 0$  flip-flop SETS on positive clock edge. (If already SET, it remains SET.)



(b)  $S = 0, R = 1$  flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)



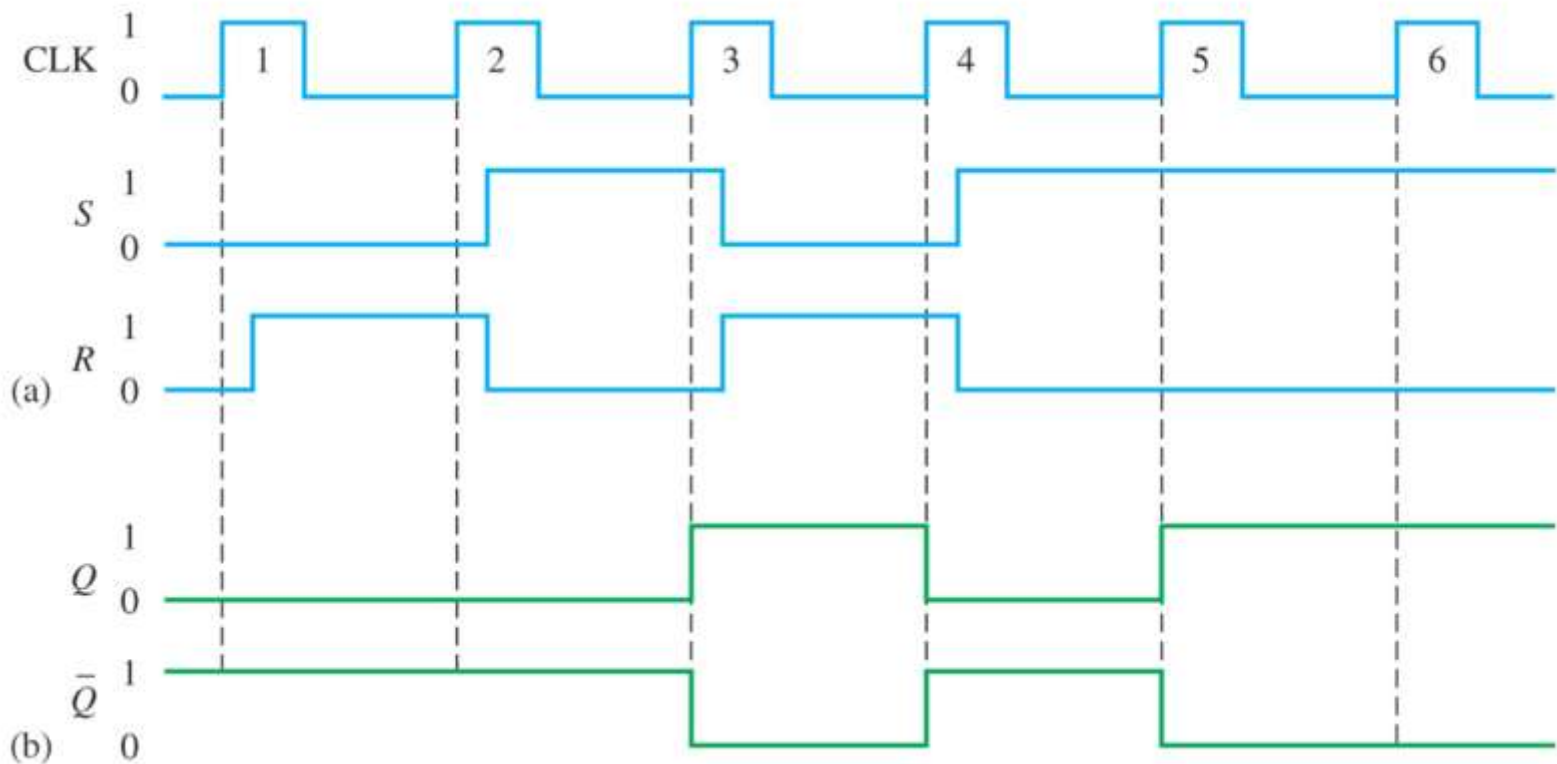
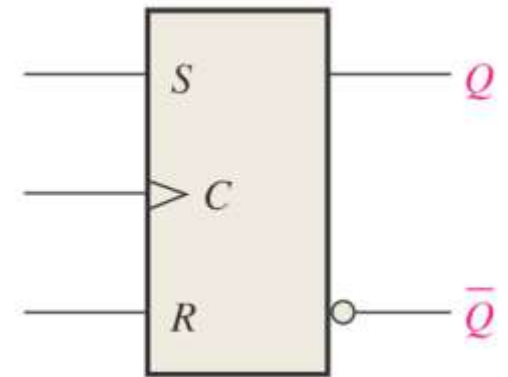
(c)  $S = 0, R = 0$  flip-flop does not change. (If SET, it remains SET; if RESET, it remains RESET.)

# Operation of a positive edge-triggered S-R flip-flop

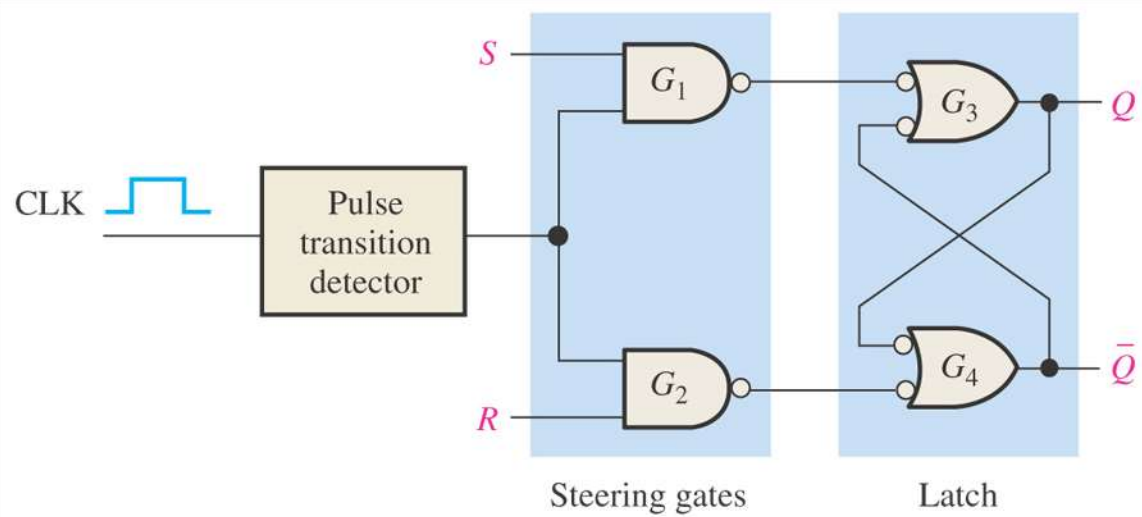
| inputs |   |            | outputs |                  |           |
|--------|---|------------|---------|------------------|-----------|
| S      | R | CLK        | Q       | $\overline{Q}$   |           |
| 0      | 0 | X          | $Q_0$   | $\overline{Q_0}$ | No change |
| 0      | 1 | $\uparrow$ | 0       | 1                | RESET     |
| 1      | 0 | $\uparrow$ | 1       | 0                | SET       |
| 1      | 1 | $\uparrow$ | ?       | ?                | invalid   |

## Example:

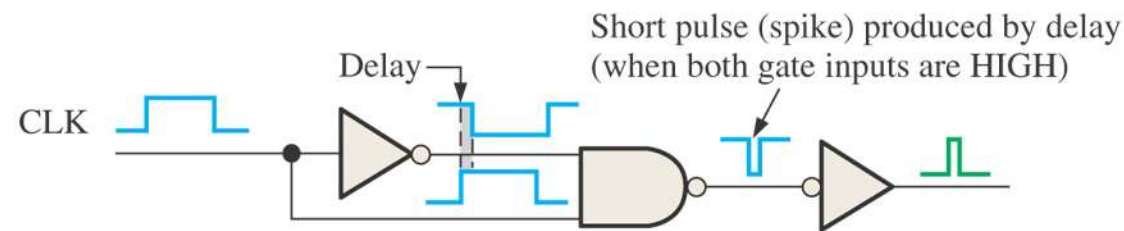
Determine the output waveforms of the flip-flop. Assume the flip-flop is initially RESET.



**Figure 7-17** Edge triggering.

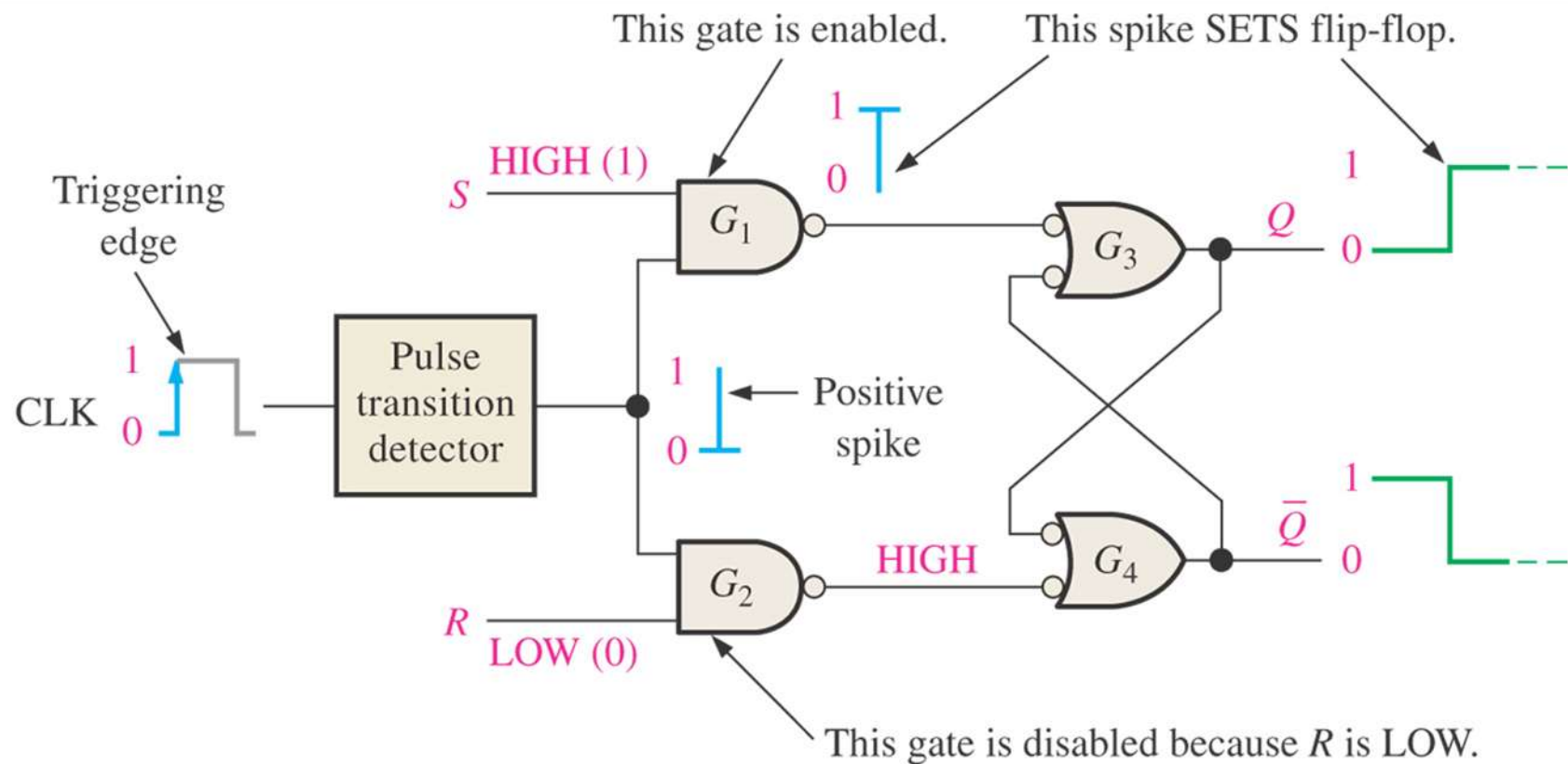


(a) A simplified logic diagram for a positive edge-triggered S-R flip-flop

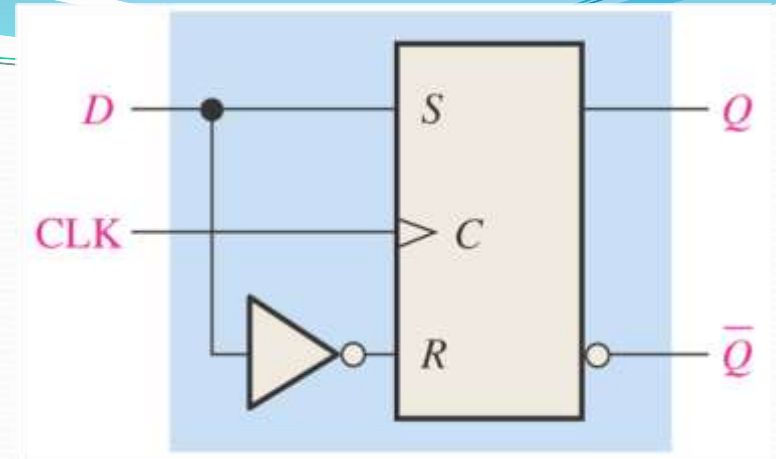


(b) A type of pulse transition detector

**Figure 7-18** Flip-flop making a transition from the RESET state to the SET state on the positive-going edge of the clock pulse.



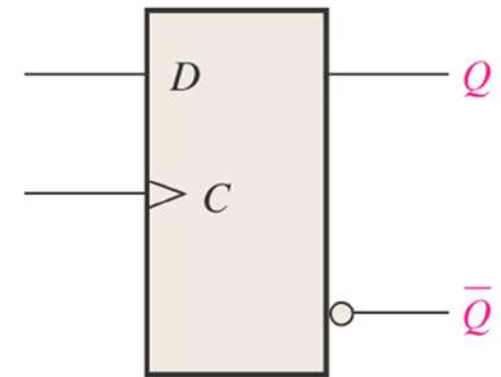
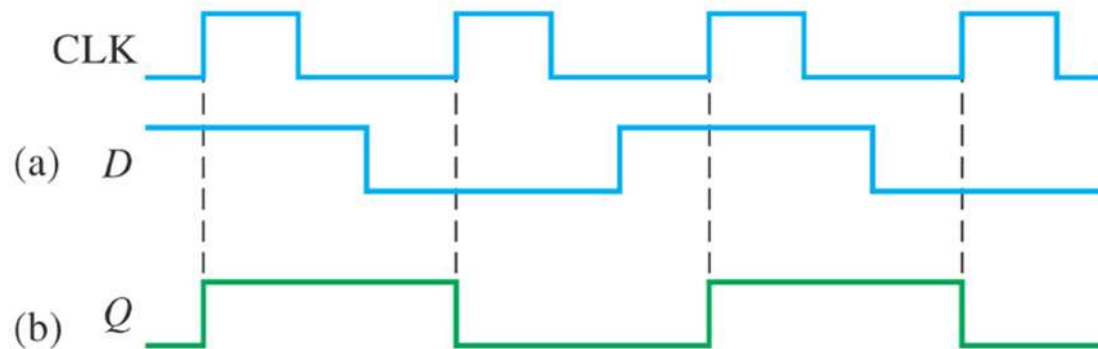
# The Edge-Triggered D Flip-Flop



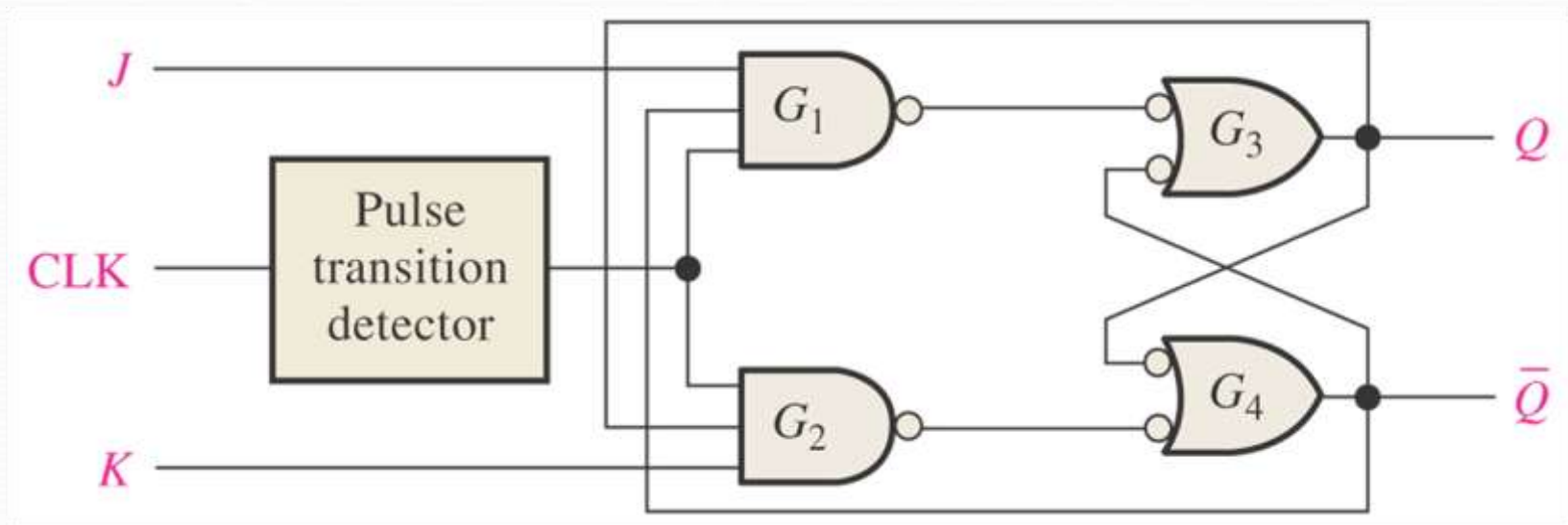
| Inputs |     | Outputs |           |       |
|--------|-----|---------|-----------|-------|
| D      | CLK | Q       | $\bar{Q}$ |       |
| 1      | ↑   | 1       | 0         | Set   |
| 0      | ↑   | 0       | 1         | RESET |

## Example:

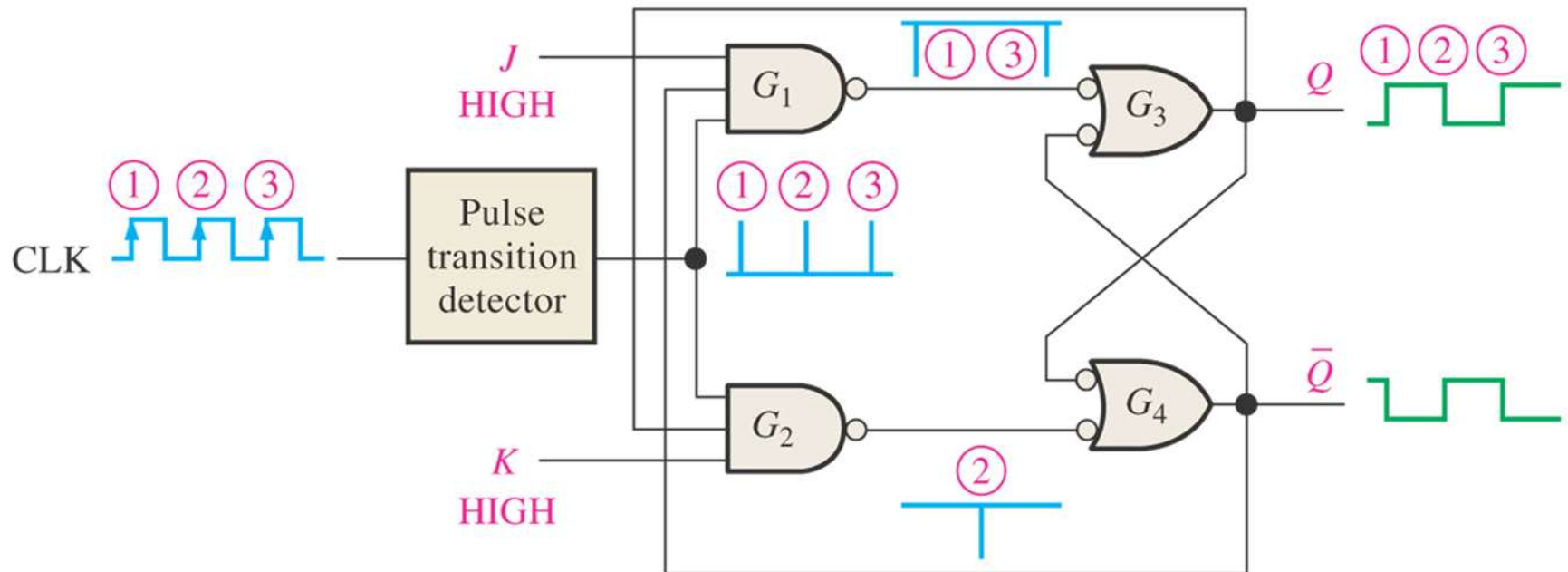
Determine the Q output waveform if the flip-flop starts out RESET given the waveforms for the D input as shown.



# The Edge-Triggered J-K Flip-Flop



**Figure 7-23** Transitions illustrating the toggle operation when  $J = 1$  and  $K = 1$ .



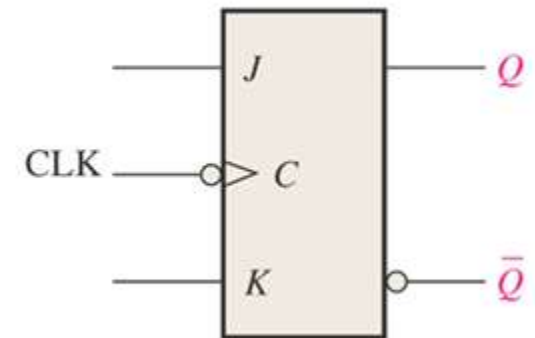
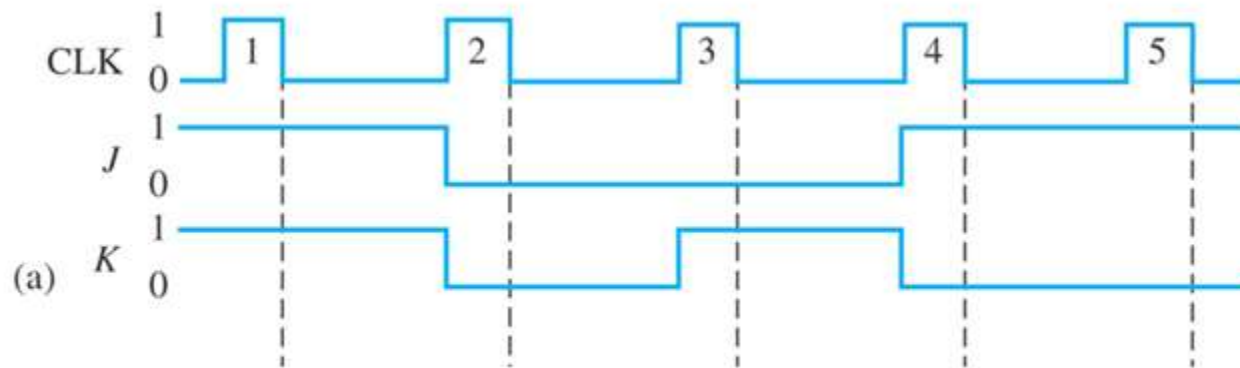
# Operation of a positive edge-triggered J-K flip-flop

| inputs |   |     | outputs          |                      |           |
|--------|---|-----|------------------|----------------------|-----------|
| J      | K | CLK | $Q^{n+1}$        | $\overline{Q^{n+1}}$ |           |
| 0      | 0 | ↑   | $Q^n$            | $\overline{Q^n}$     | No change |
| 0      | 1 | ↑   | 0                | 1                    | RESET     |
| 1      | 0 | ↑   | 1                | 0                    | SET       |
| 1      | 1 | ↑   | $\overline{Q^n}$ | $Q^n$                | Toggle    |

$$Q^{n+1} = J\overline{Q^n} + \overline{K}Q^n$$

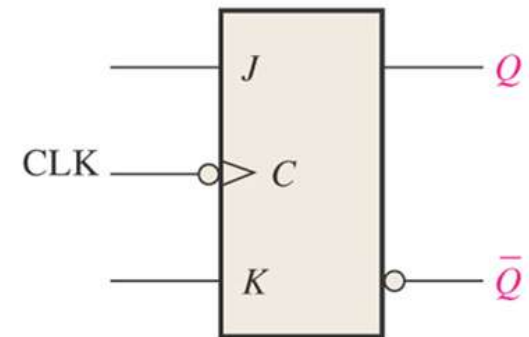
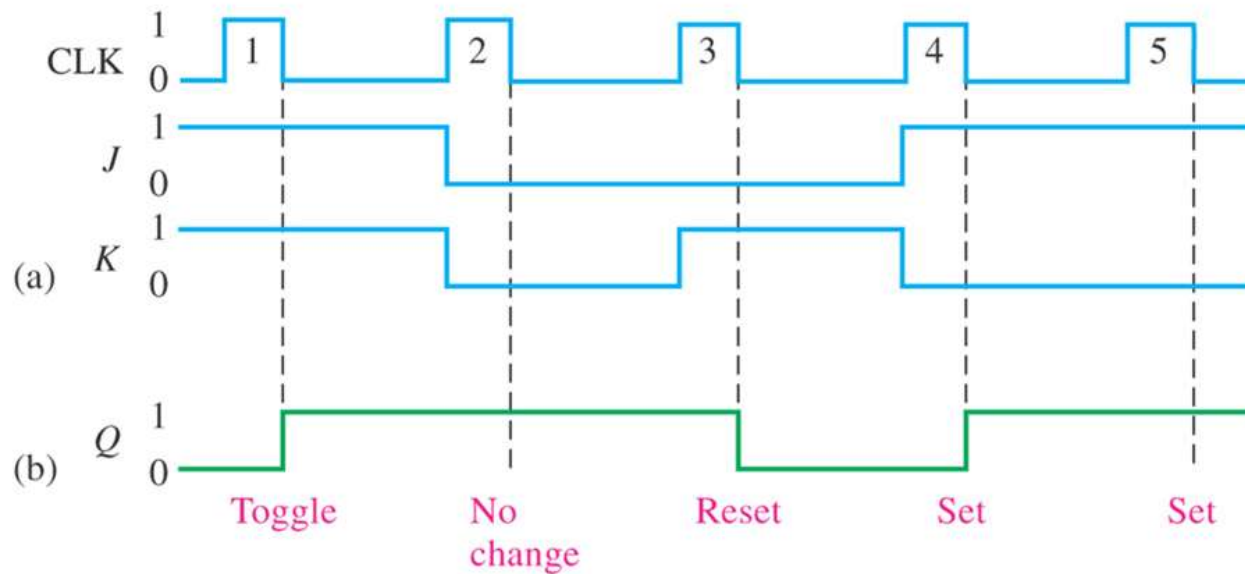
## Example

Determine the Q output, assuming the flip-flop is initially RESET.



## Example

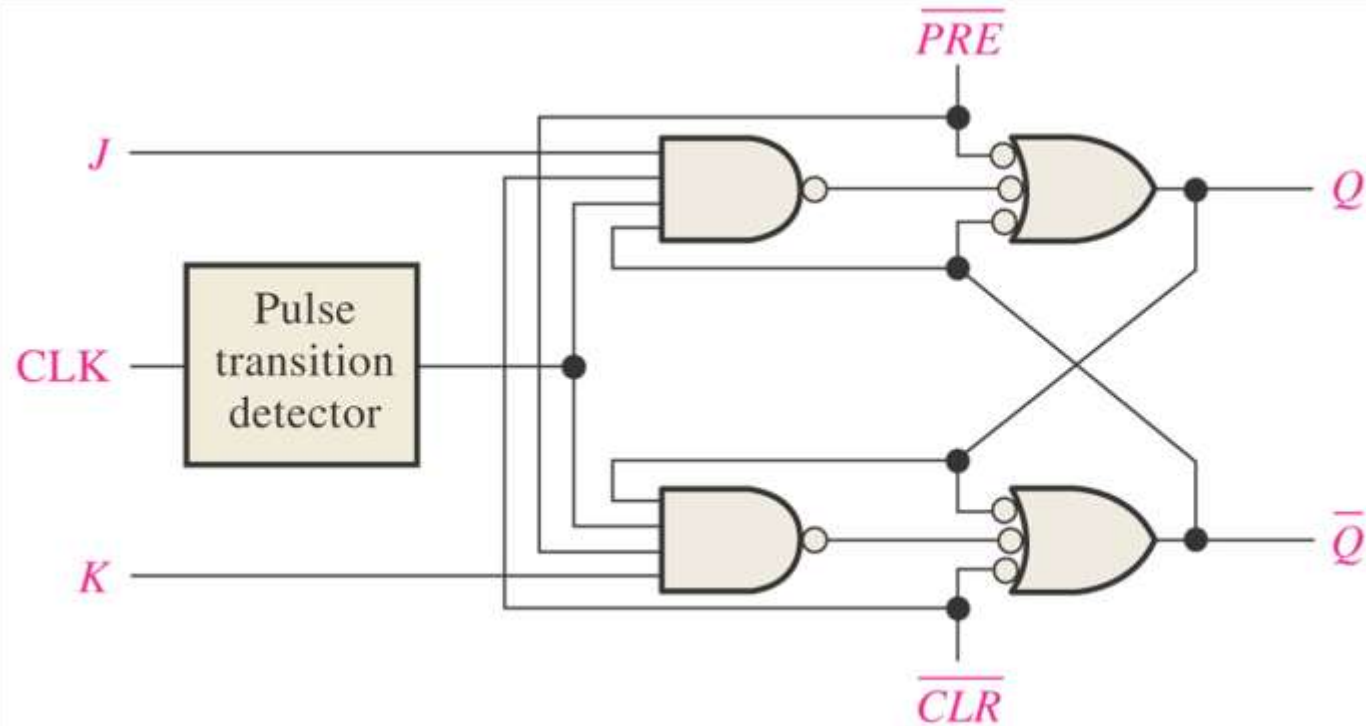
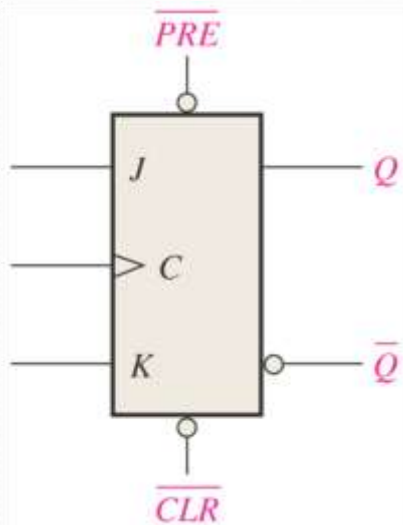
Determine the Q output, assuming the flip-flop is initially RESET.



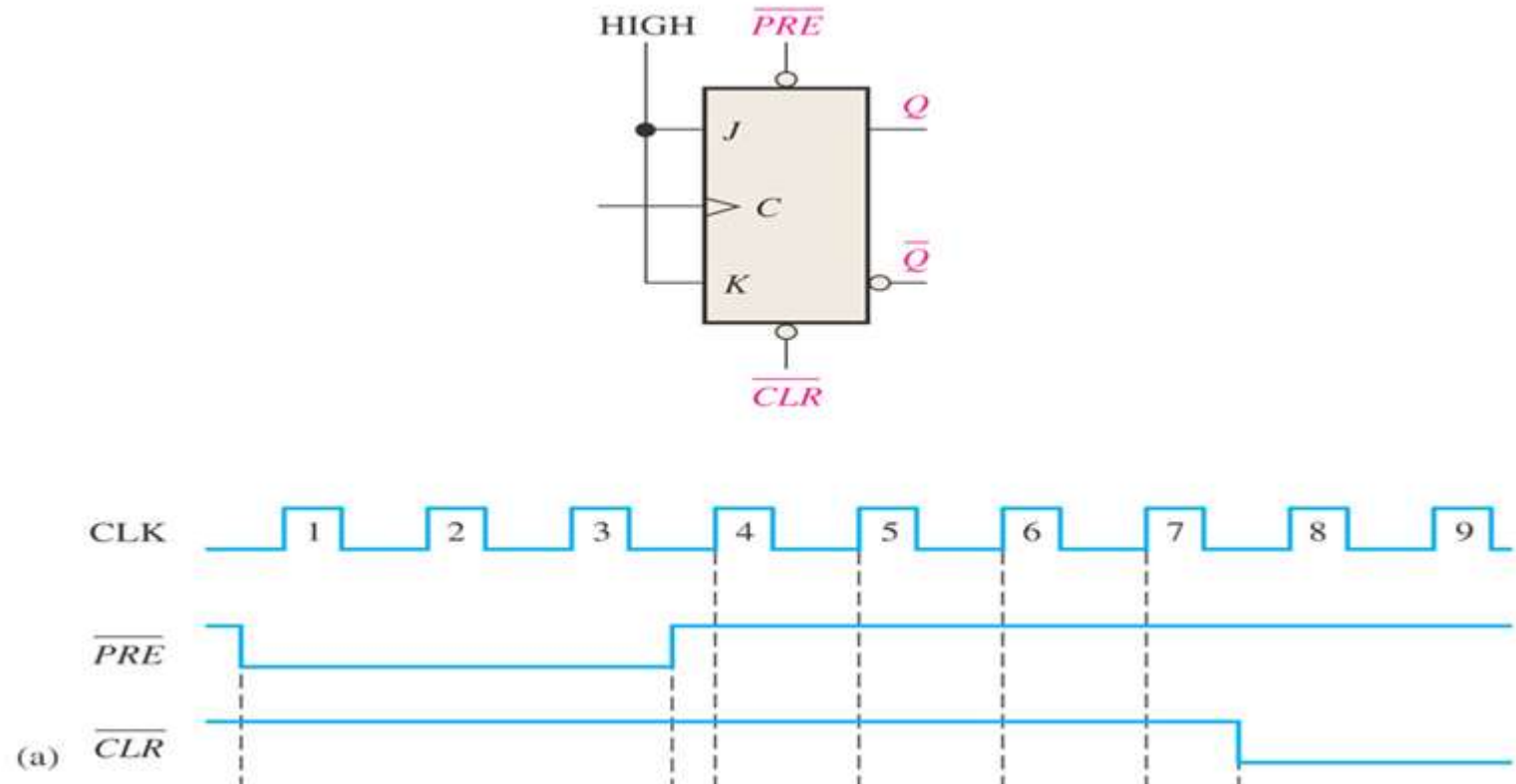
# Asynchronous Preset and Clear Inputs

- Synchronous inputs
  - Data on inputs are transferred to the flip-flop's output only on the triggering edge of the clock pulse
- Asynchronous inputs
  - Inputs affect the state of the flip-flop independent of the clock
  - Preset, CLEAR, direct set, direct reset

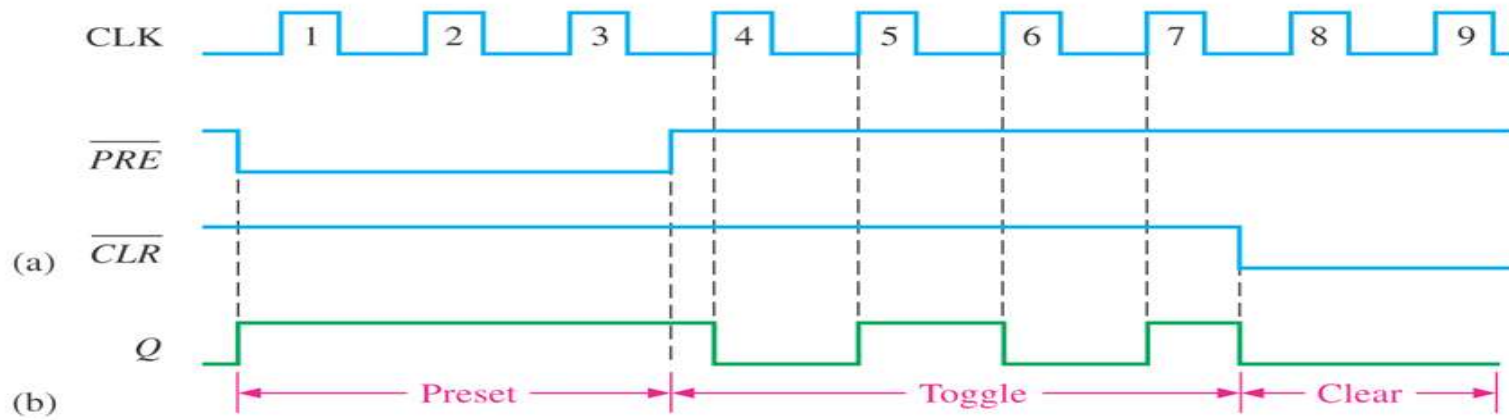
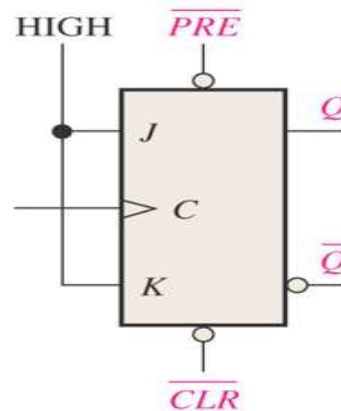
# Logic symbol and logic diagram for a basic J-K flip-flop



Example: Determine the Q output for the inputs shown in the timing diagram if Q is initially LOW.

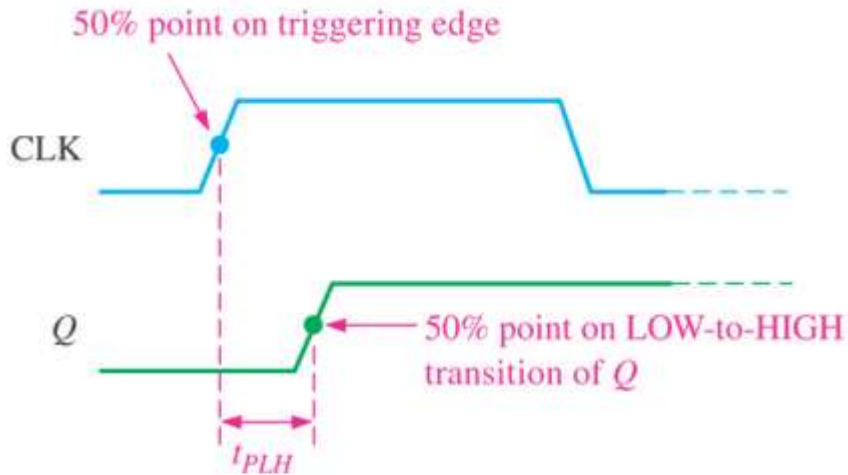


Example: Determine the Q output for the inputs shown in the timing diagram if Q is initially LOW.

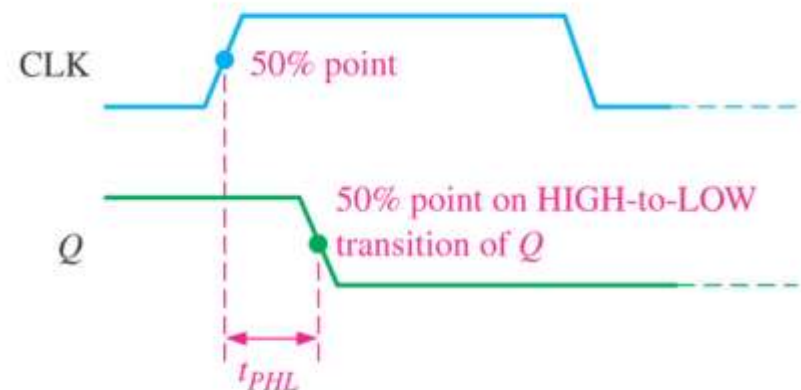


## 7.3 Flip-Flop Operating Characteristics

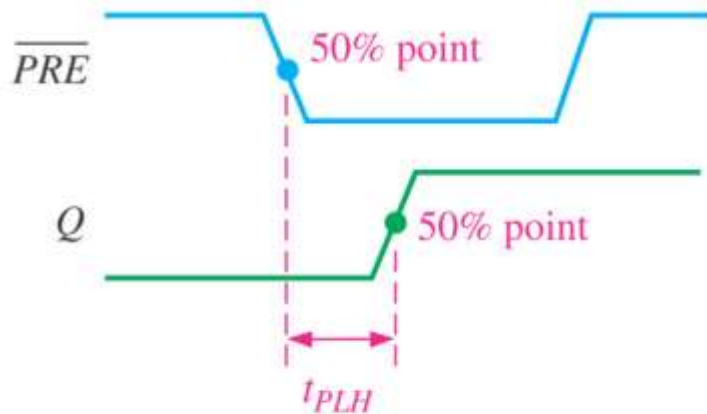
# Propagation Delay Times



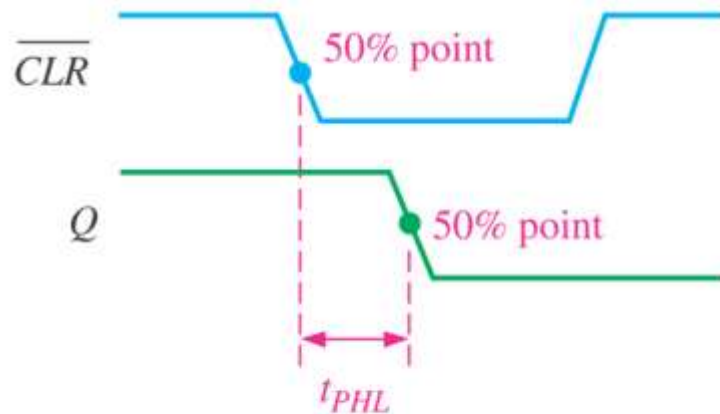
(a)



(b)

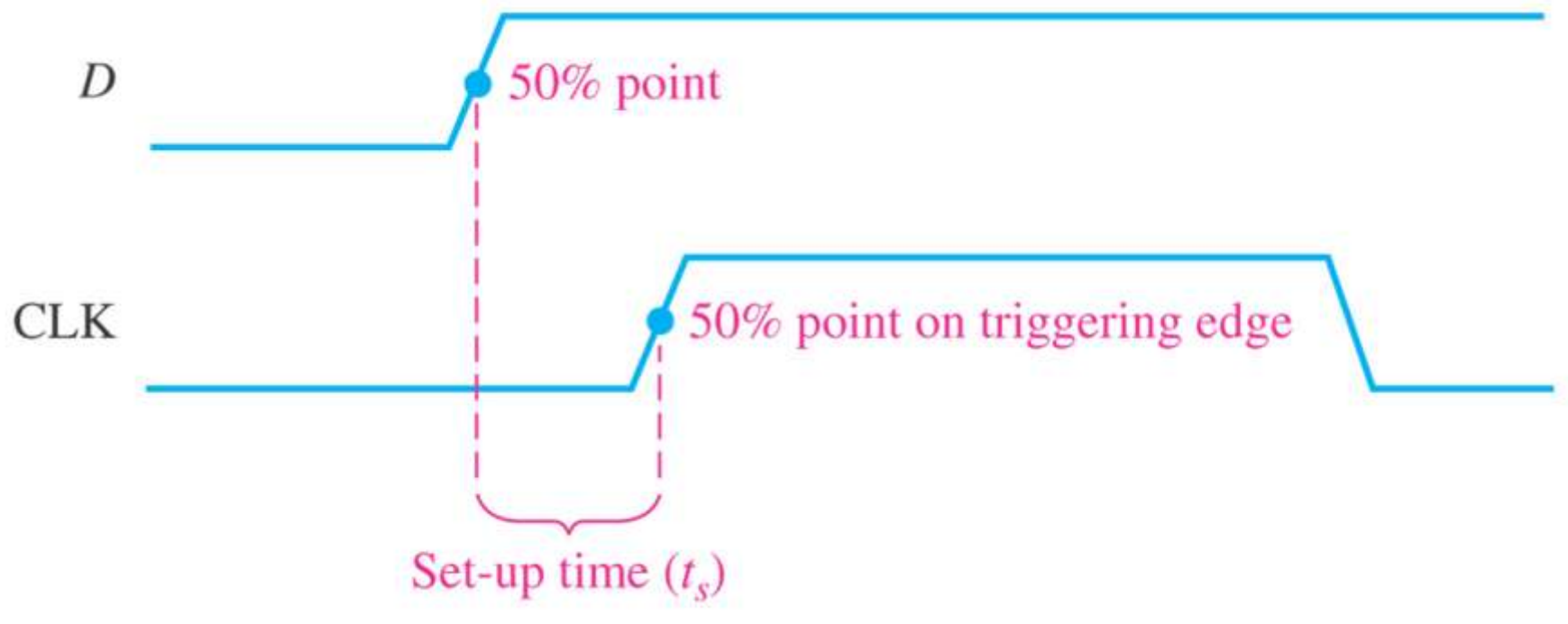


(a)



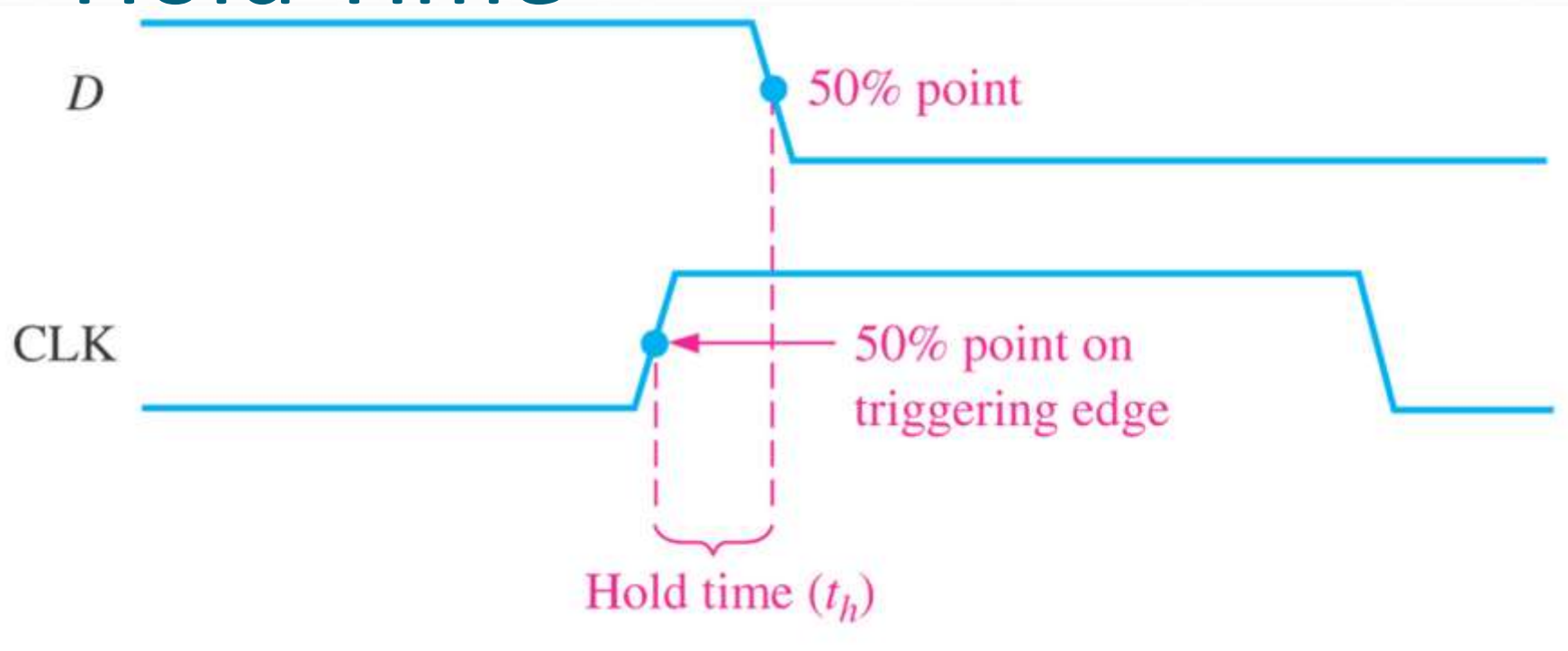
(b)

# Set-up Time



**Figure 7–34** Set-up time ( $t_s$ ). The logic level must be present on the *D* input for a time equal to or greater than  $t_s$  before the triggering edge of the clock pulse for reliable data entry.

# Hold Time



**Figure 7–35** Hold time ( $t_h$ ). The logic level must remain on the D input for a time equal to or greater than  $t_h$  after the triggering edge of the clock pulse for reliable data entry.

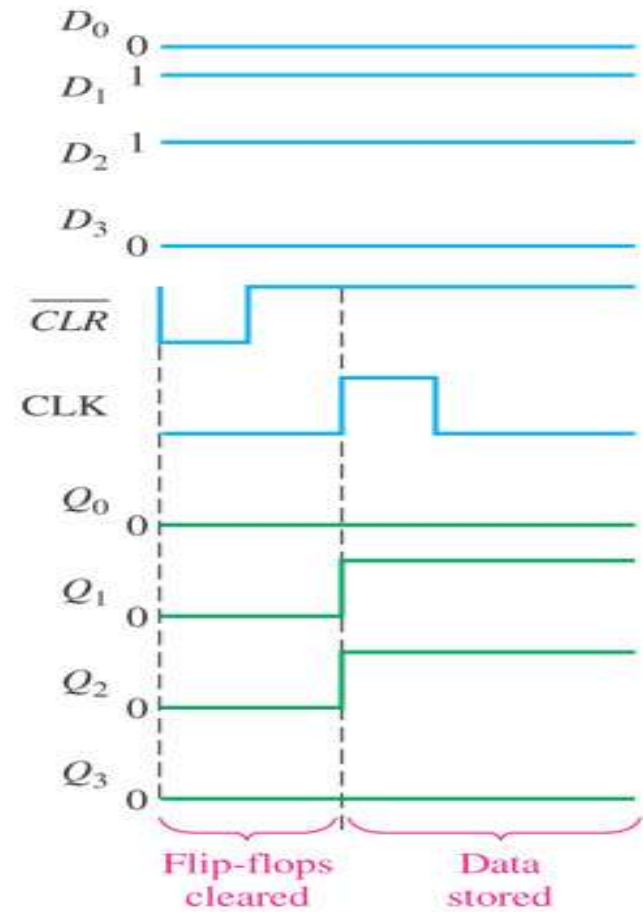
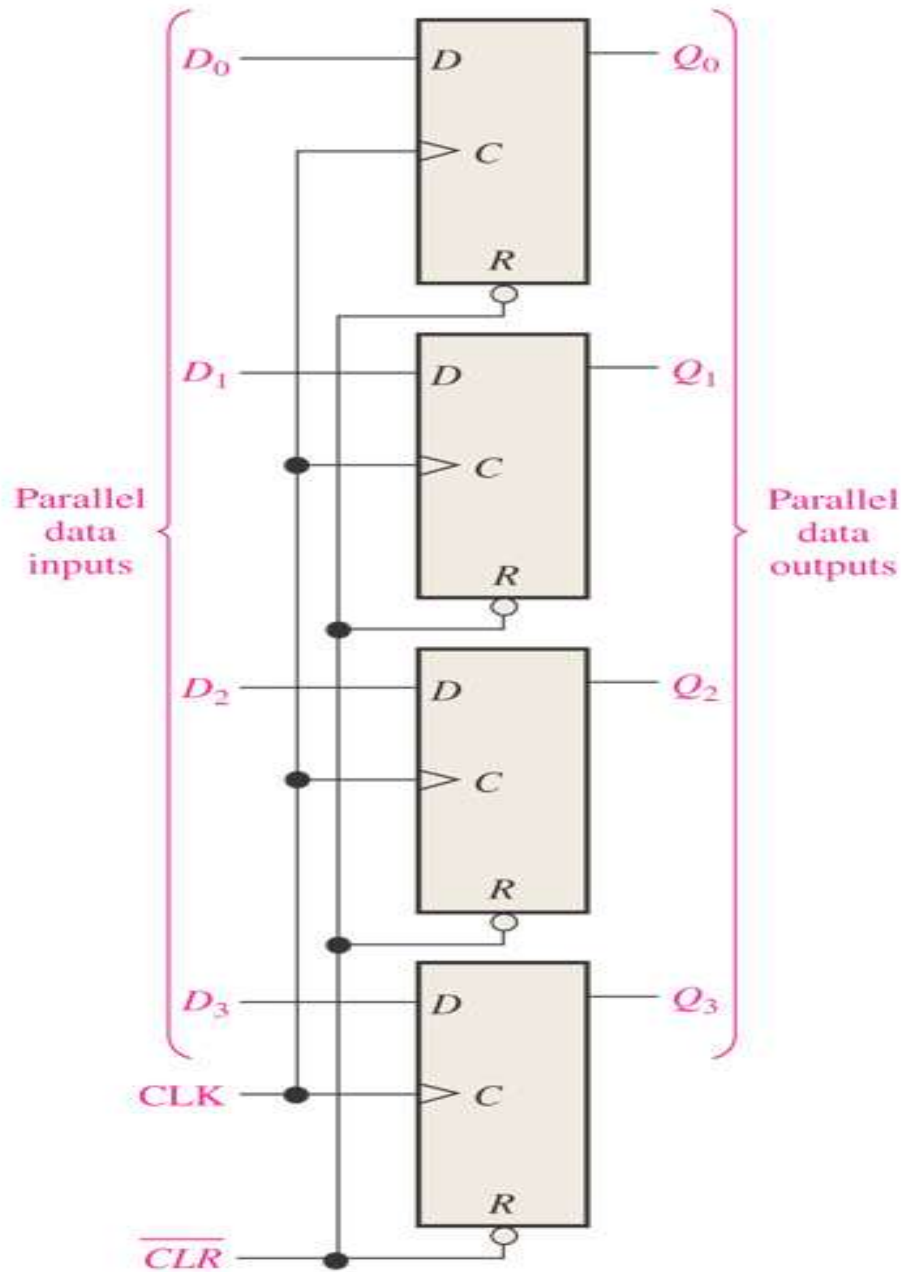
# Other Characteristics

- Maximum Clock Frequency
- Pulse Widths
- Power Dissipation

# 7.4 Flip-Flop Applications

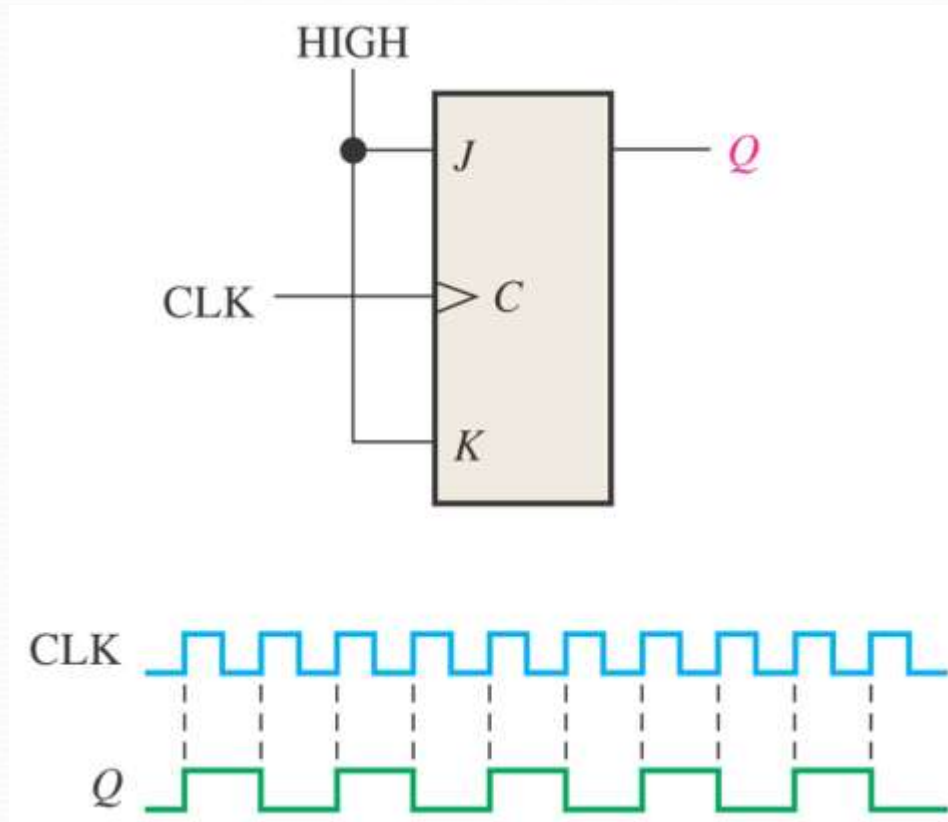
- Parallel Data Storage
- Frequency division
- Counting

# Parallel Data Storage

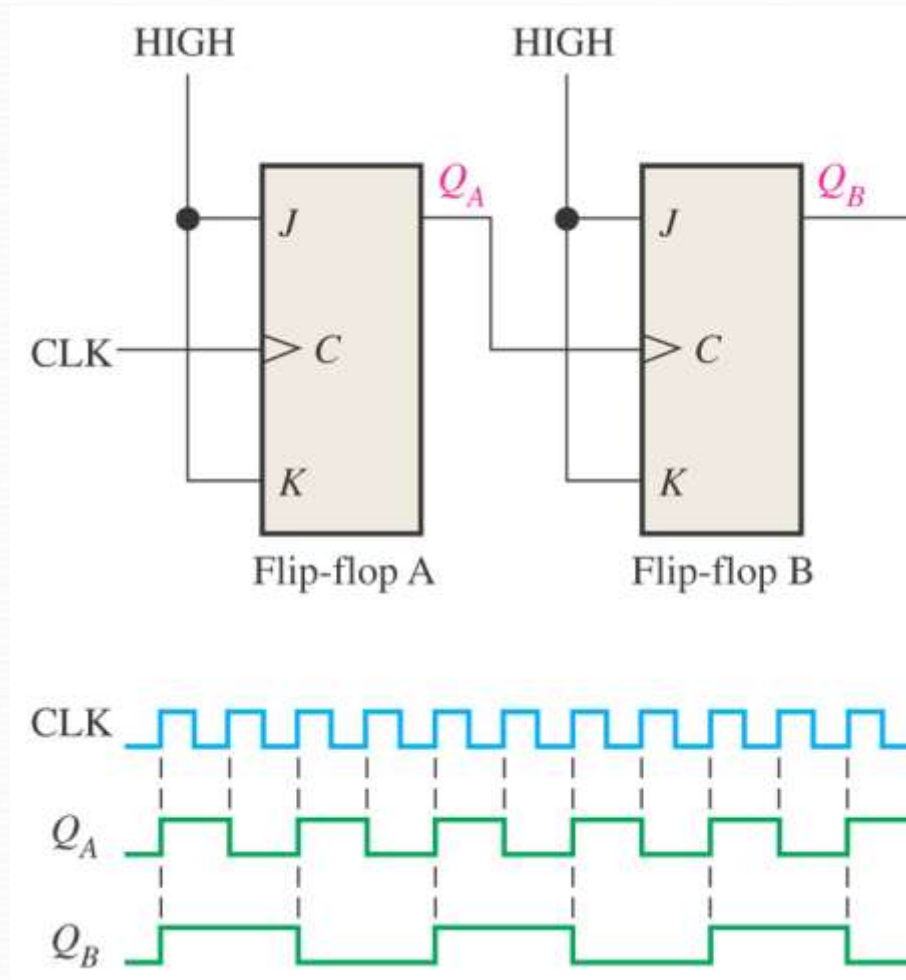


(b)

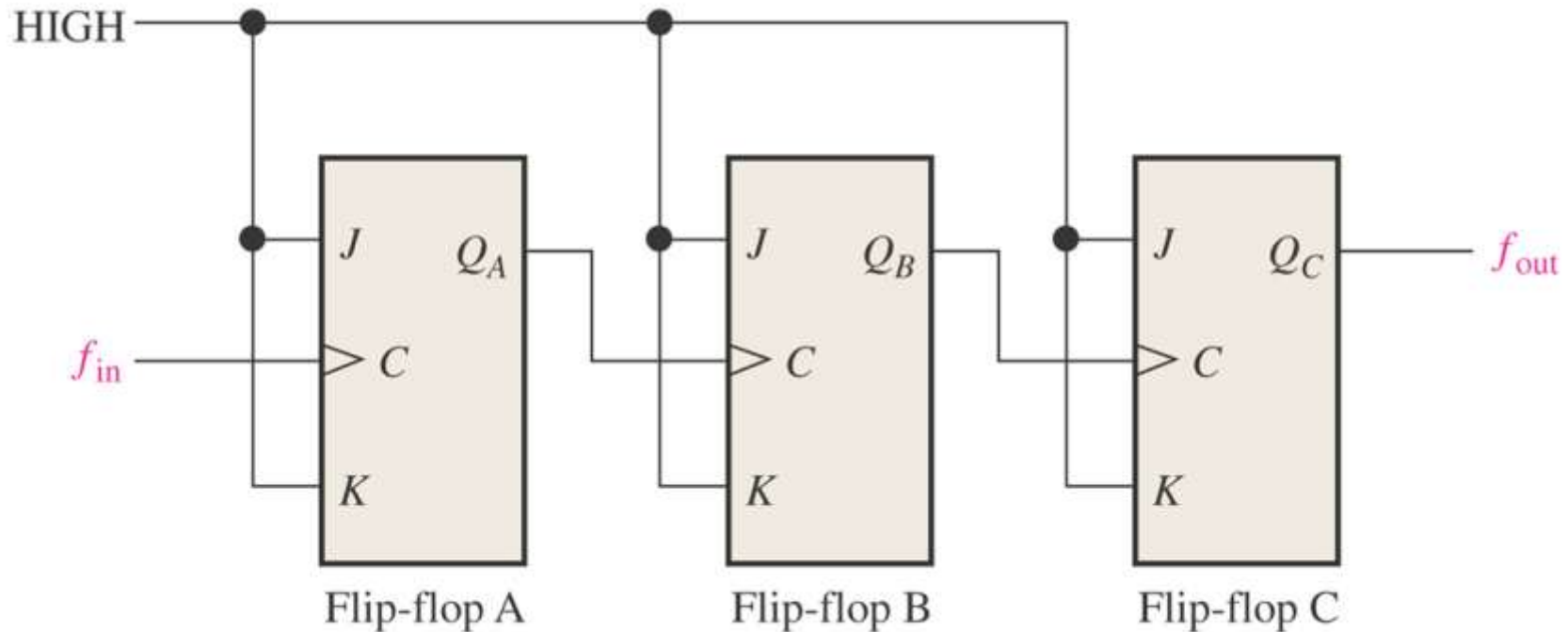
- Frequency Division

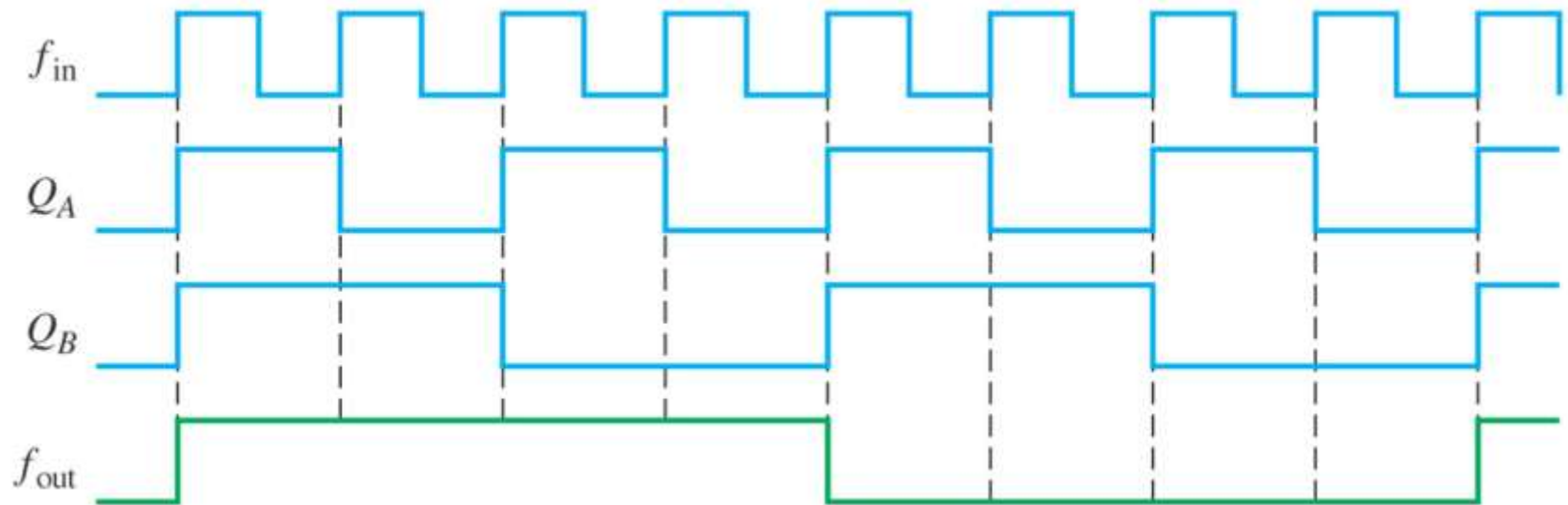


**Figure 7–38** Example of two J-K flip-flops used to divide the clock frequency by 4.  $Q_A$  is one-half and  $Q_B$  is one-fourth the frequency of CLK.

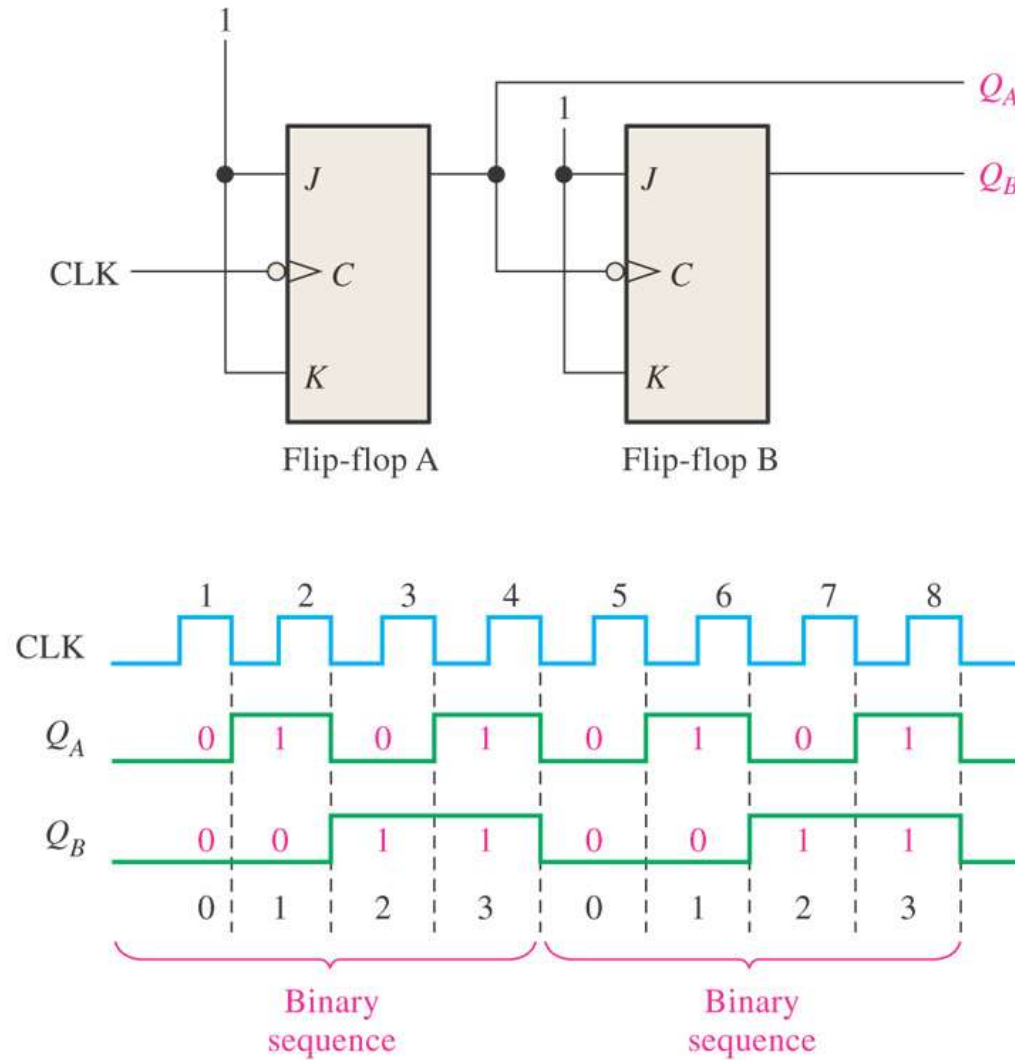


**Example:** Develop the waveform of  $Q_C$  for the circuit shown below when an 8KHz square wave input is applied to the clock input of flip-flop A.





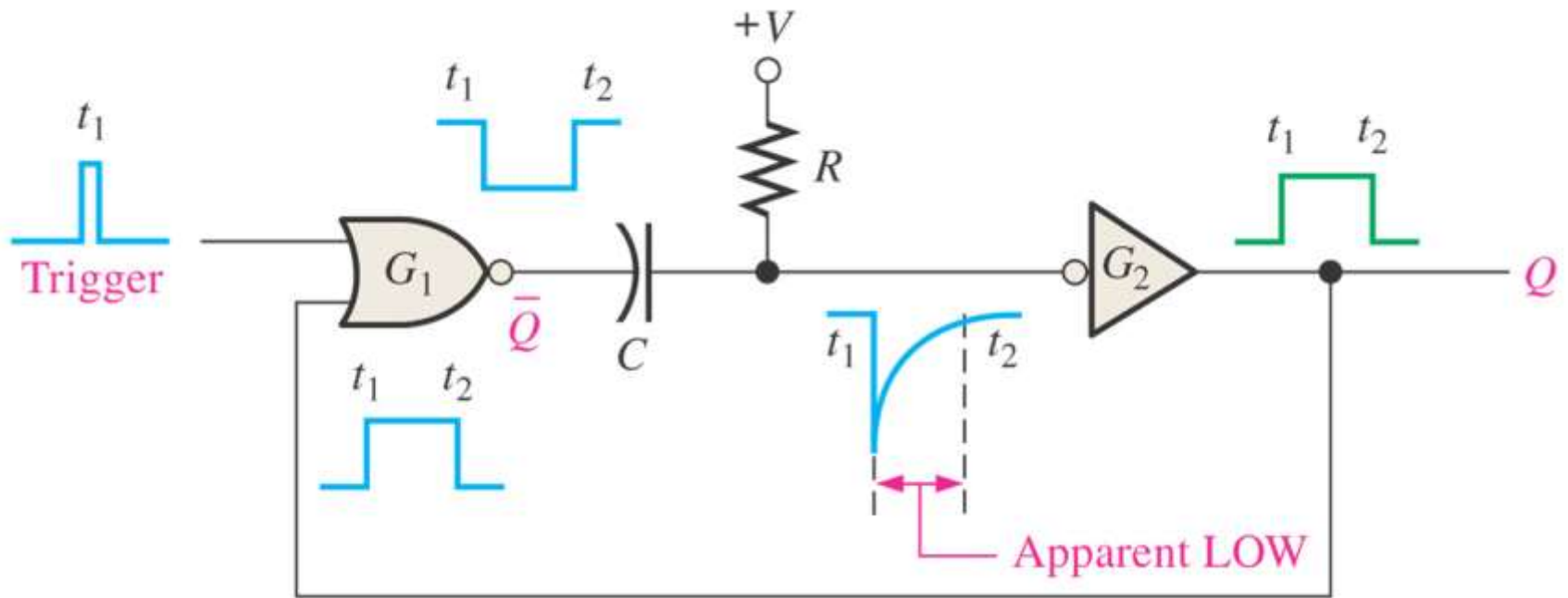
# Counting



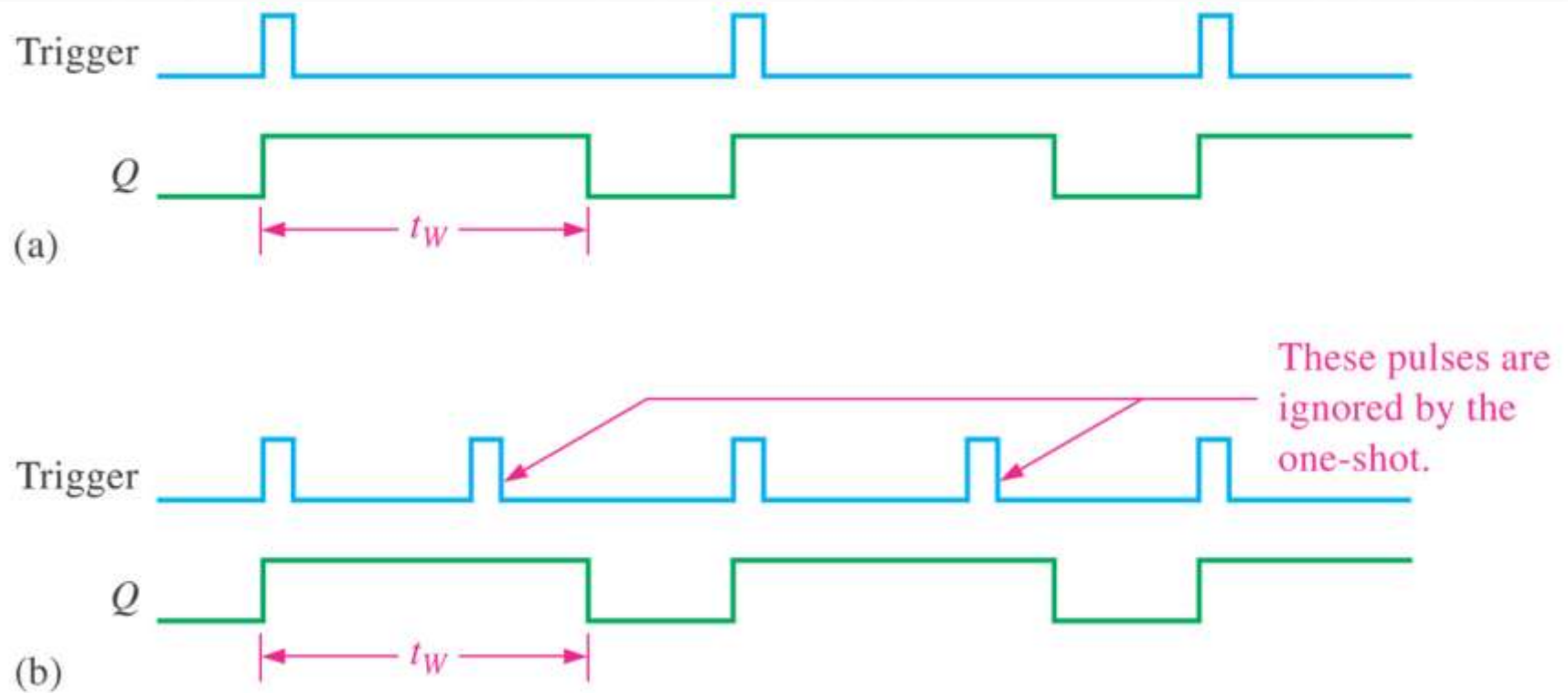
## 7.5 One-Shots (单稳态触发器)

- A device with only one stable state
- Normally in its stable state and will change to its unstable state only when triggered
- Once be triggered, it remains in its unstable state for a predetermined length of time before automatically return to its stable state

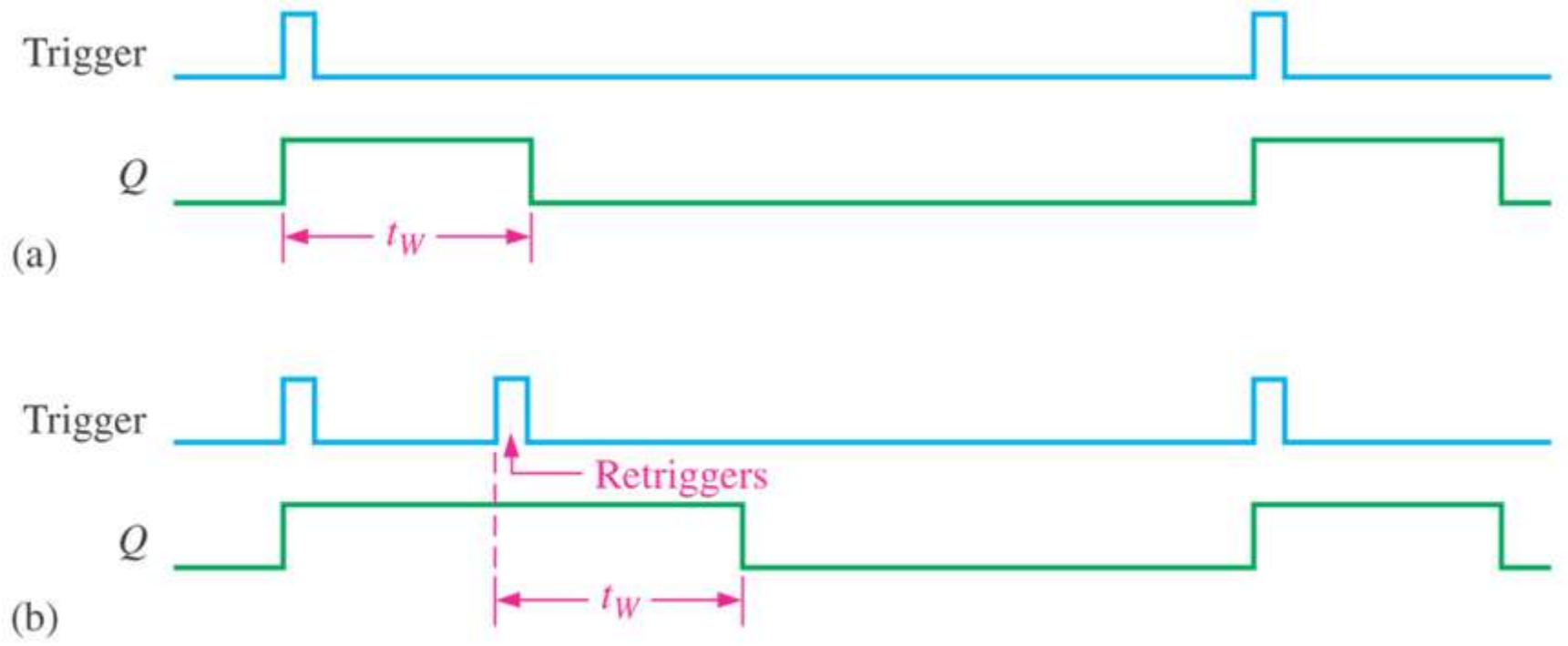
**Figure 7–44** A simple one-shot circuit.



**Figure 7–46** Non retriggerable one-shot action.

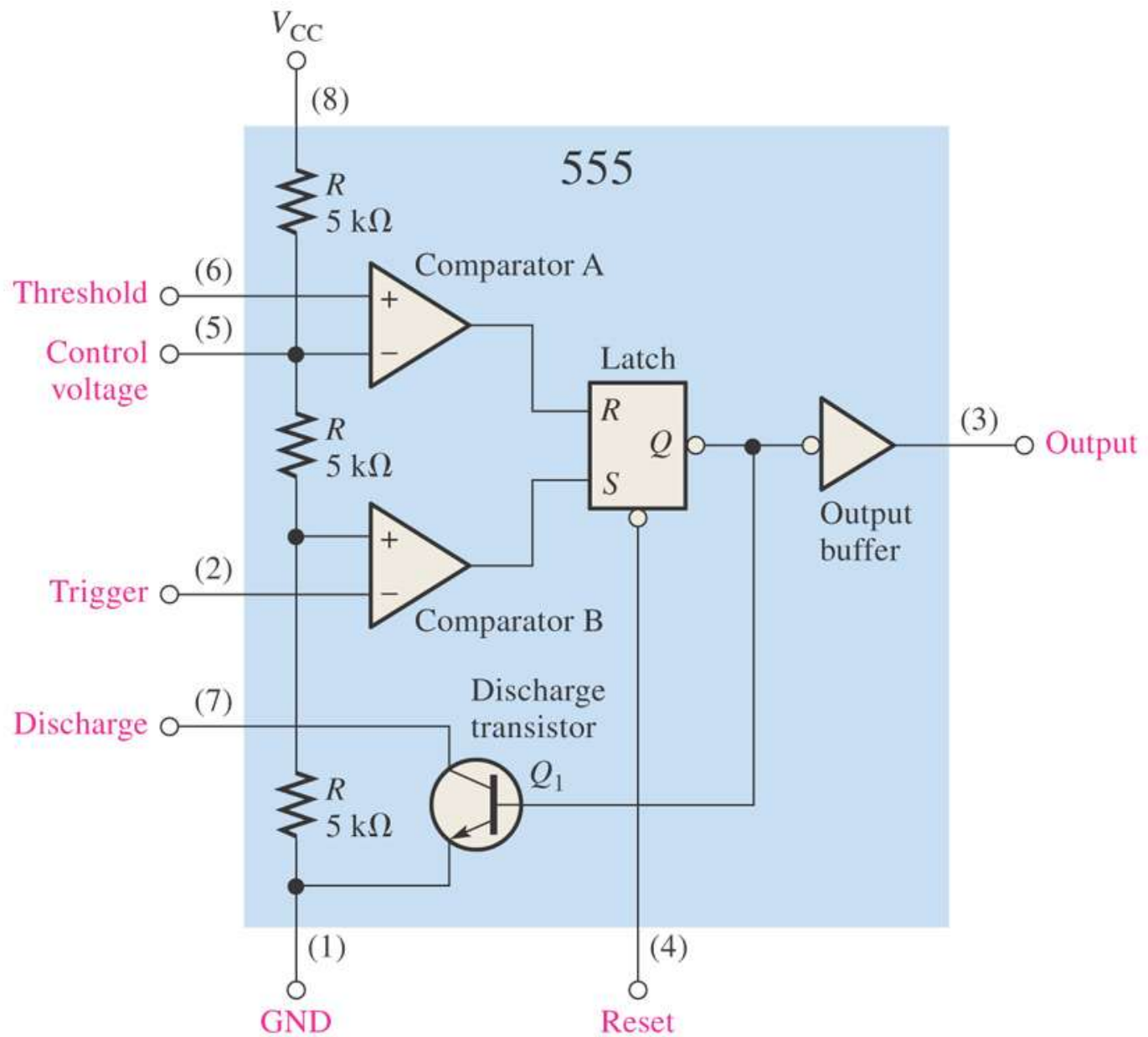


**Figure 7–47** Retriggerable one-shot action.



# 7.6 The 555 Timer

- A versatile and widely used IC device
- Can be configured in two different modes
  - A mono-stable one-shot (单稳态触发器)
  - An astable multi-vibration oscillator (非稳态多谐振荡器)



**Figure 7–53** Internal functional diagram of a 555 timer (pin numbers are in parenthesis).

**Figure 7–54** The 555 timer connected as a one-shot.

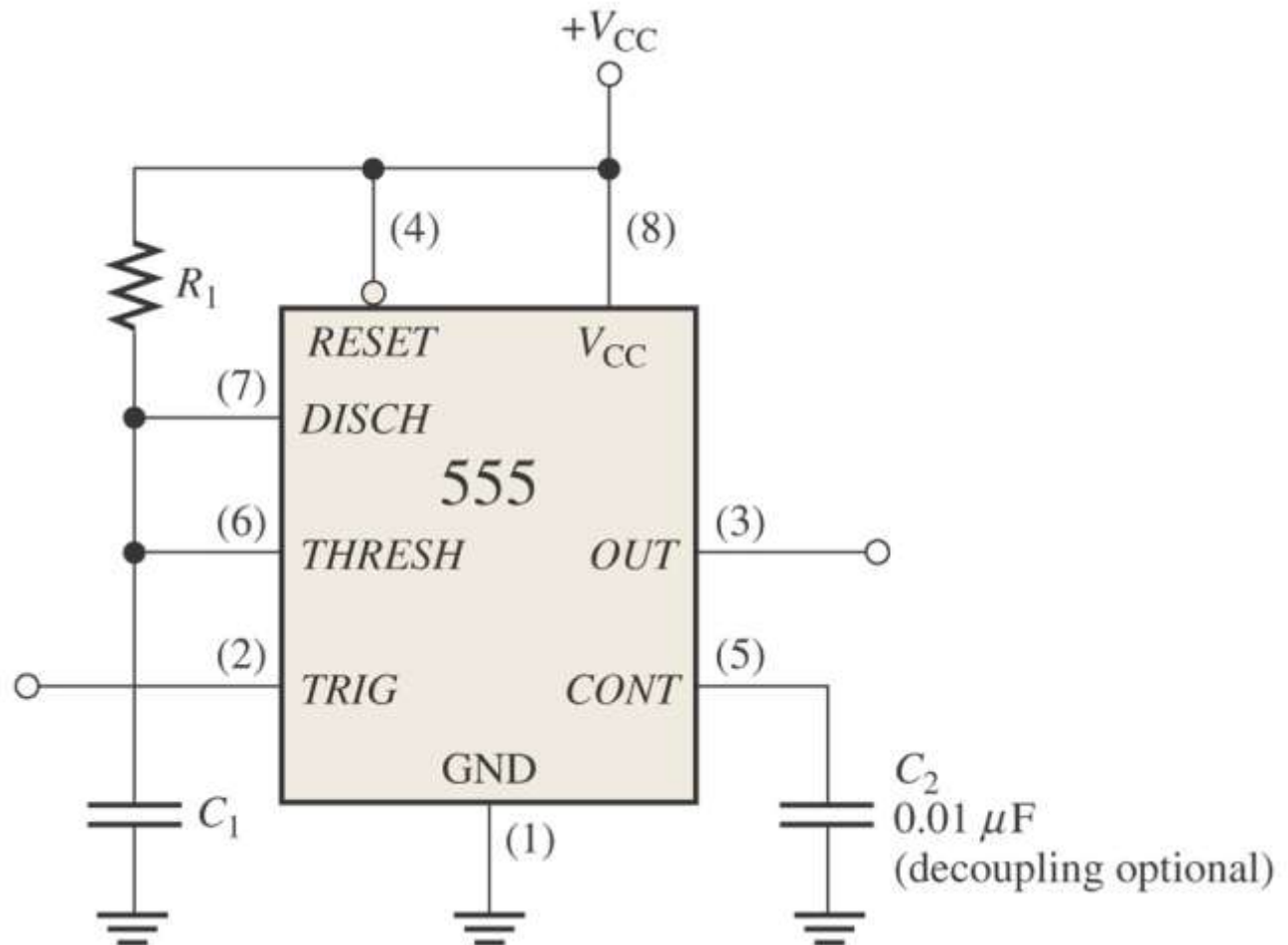
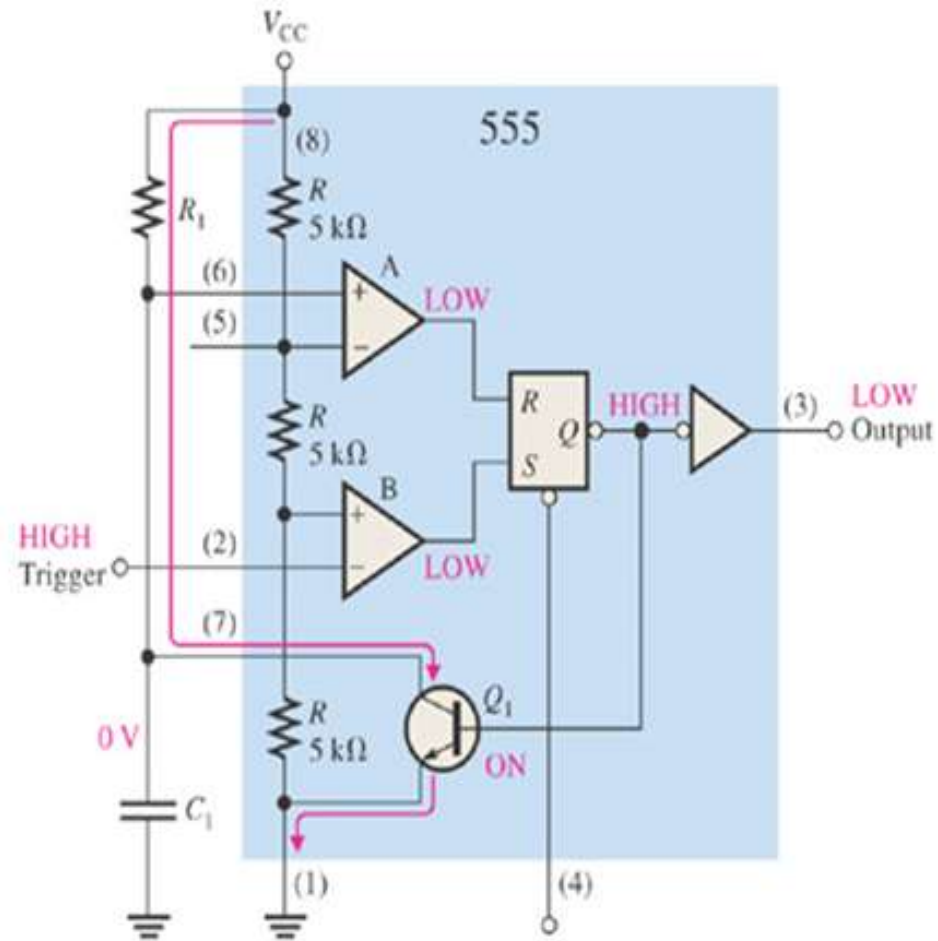
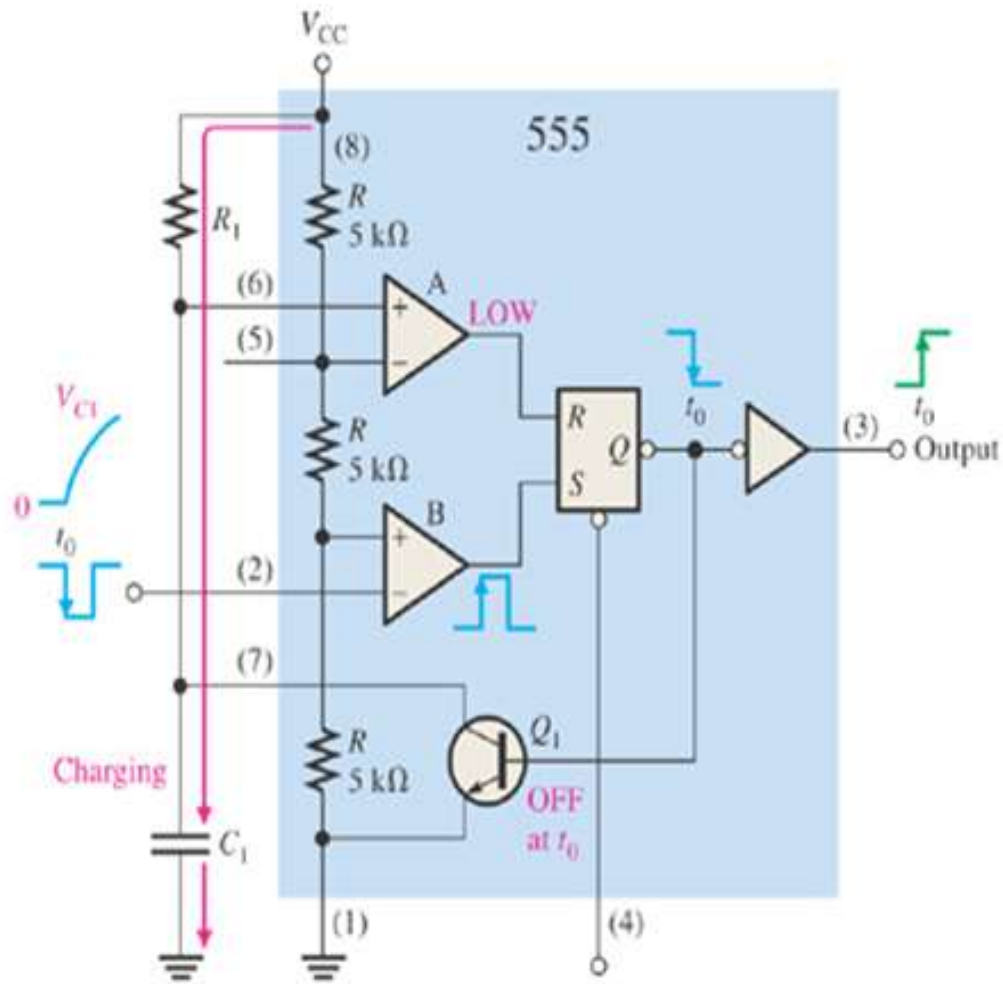


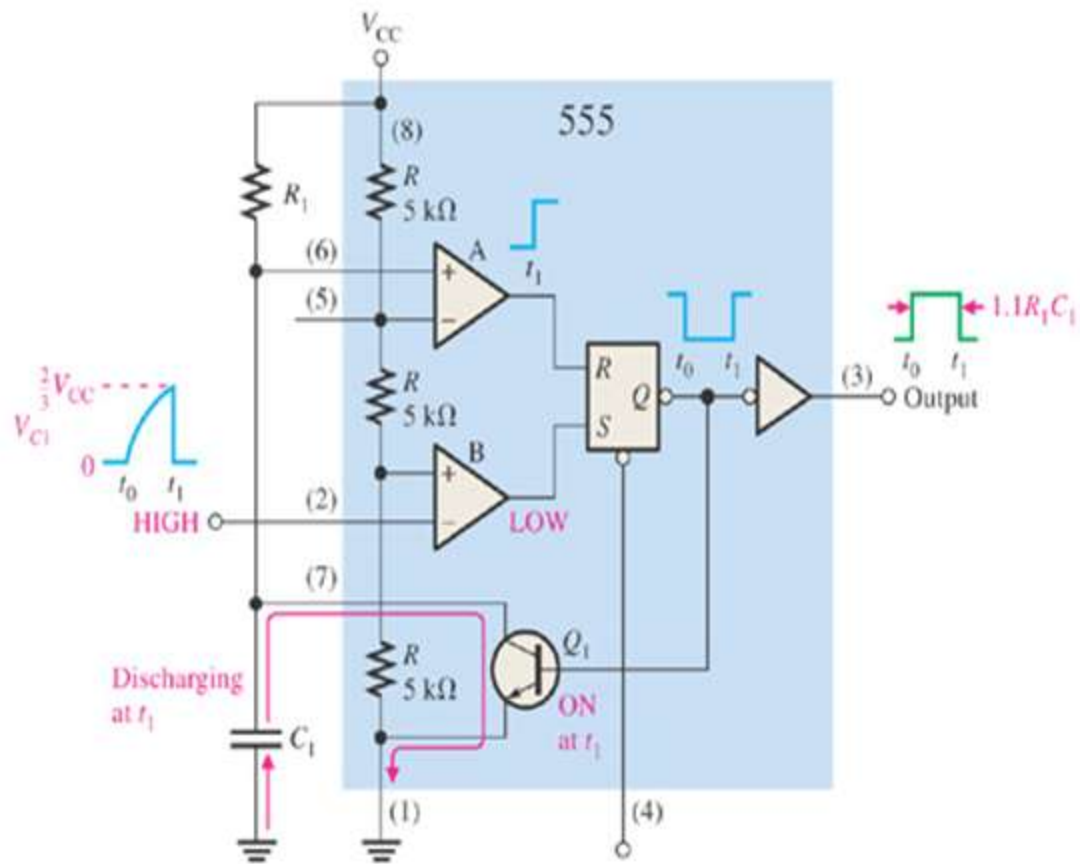
Figure 7–55 One-shot operation of the 555 timer.



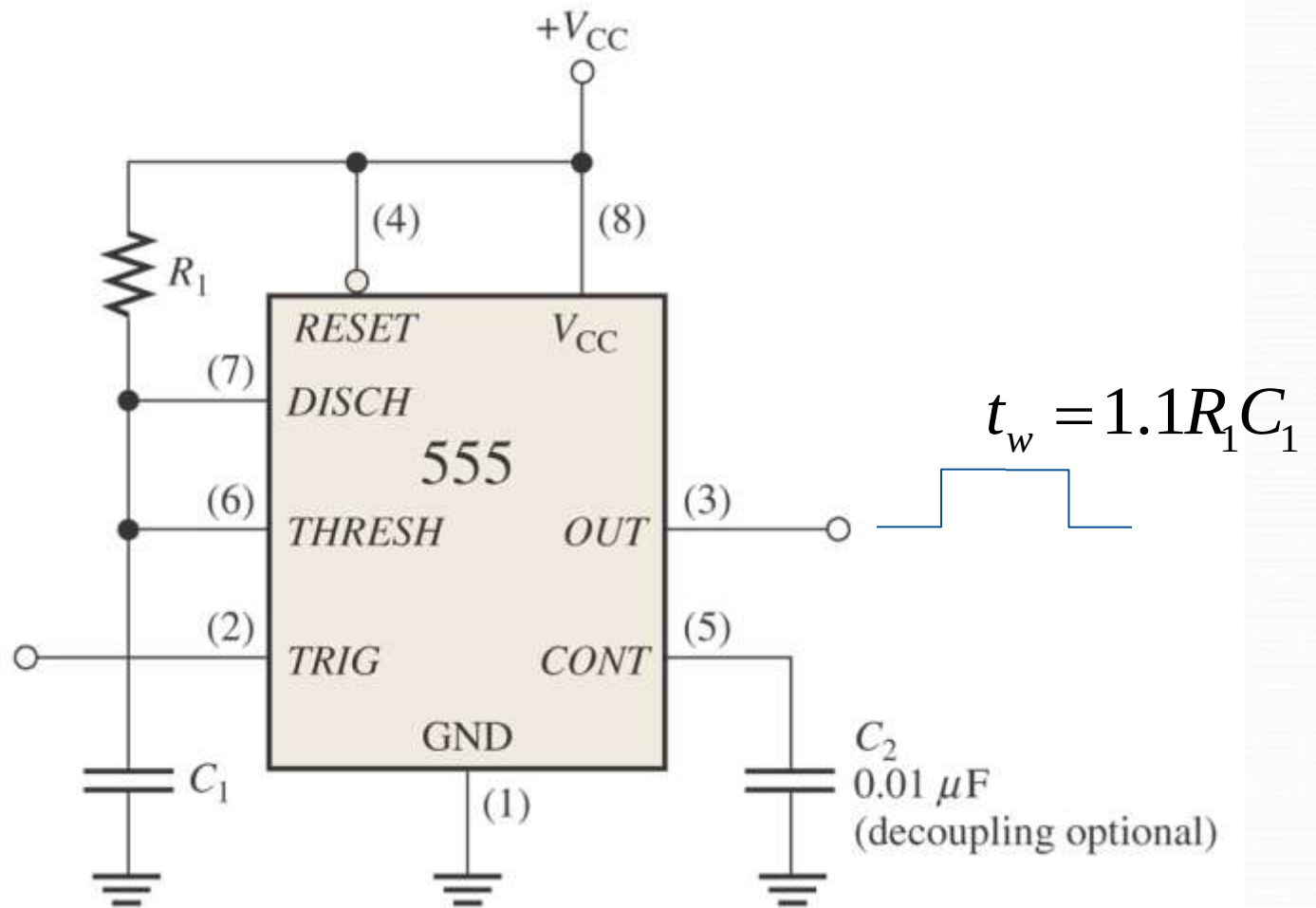
(a) Prior to triggering. (The current path is indicated by the red arrow.)



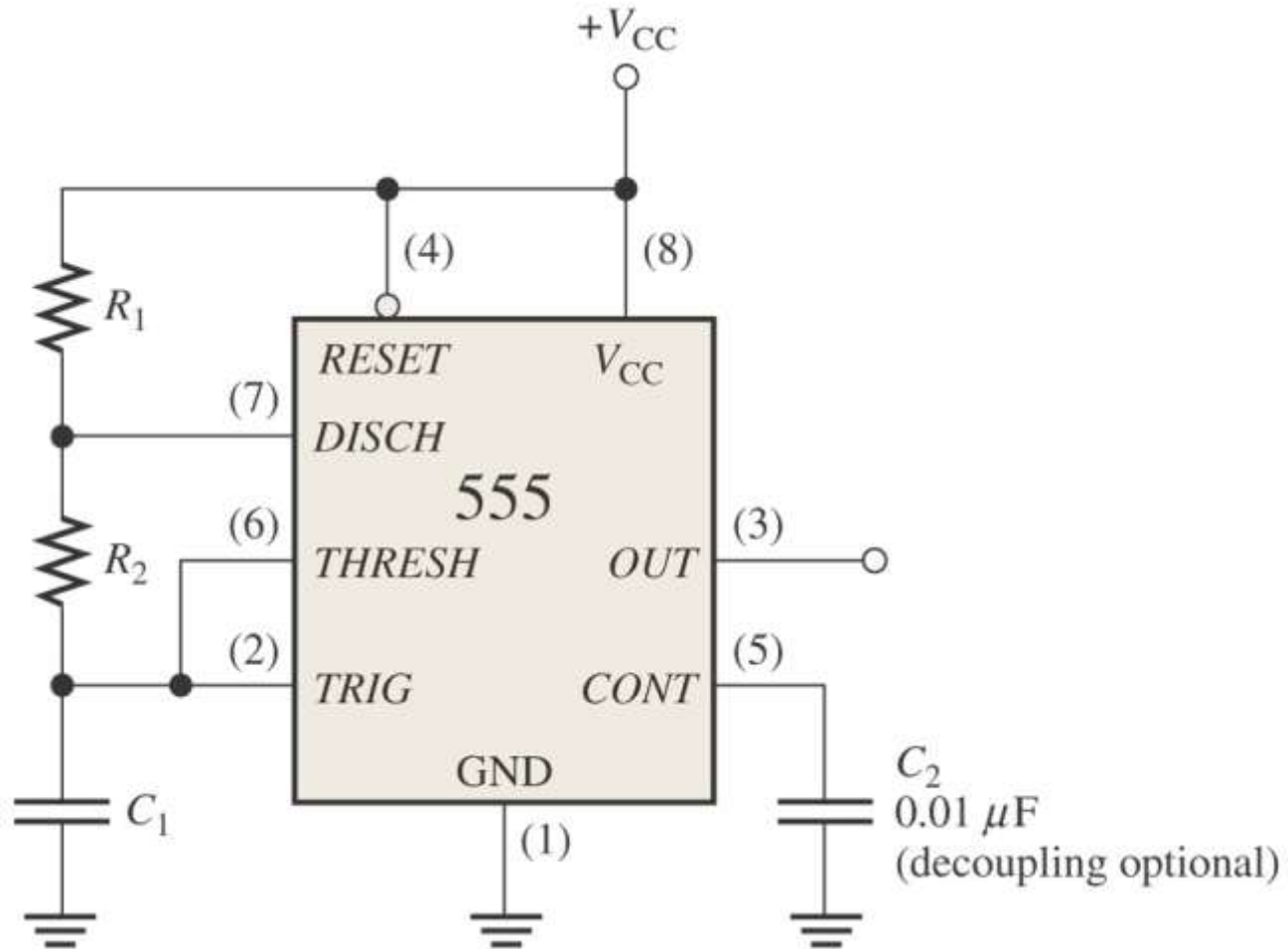
(b) When triggered



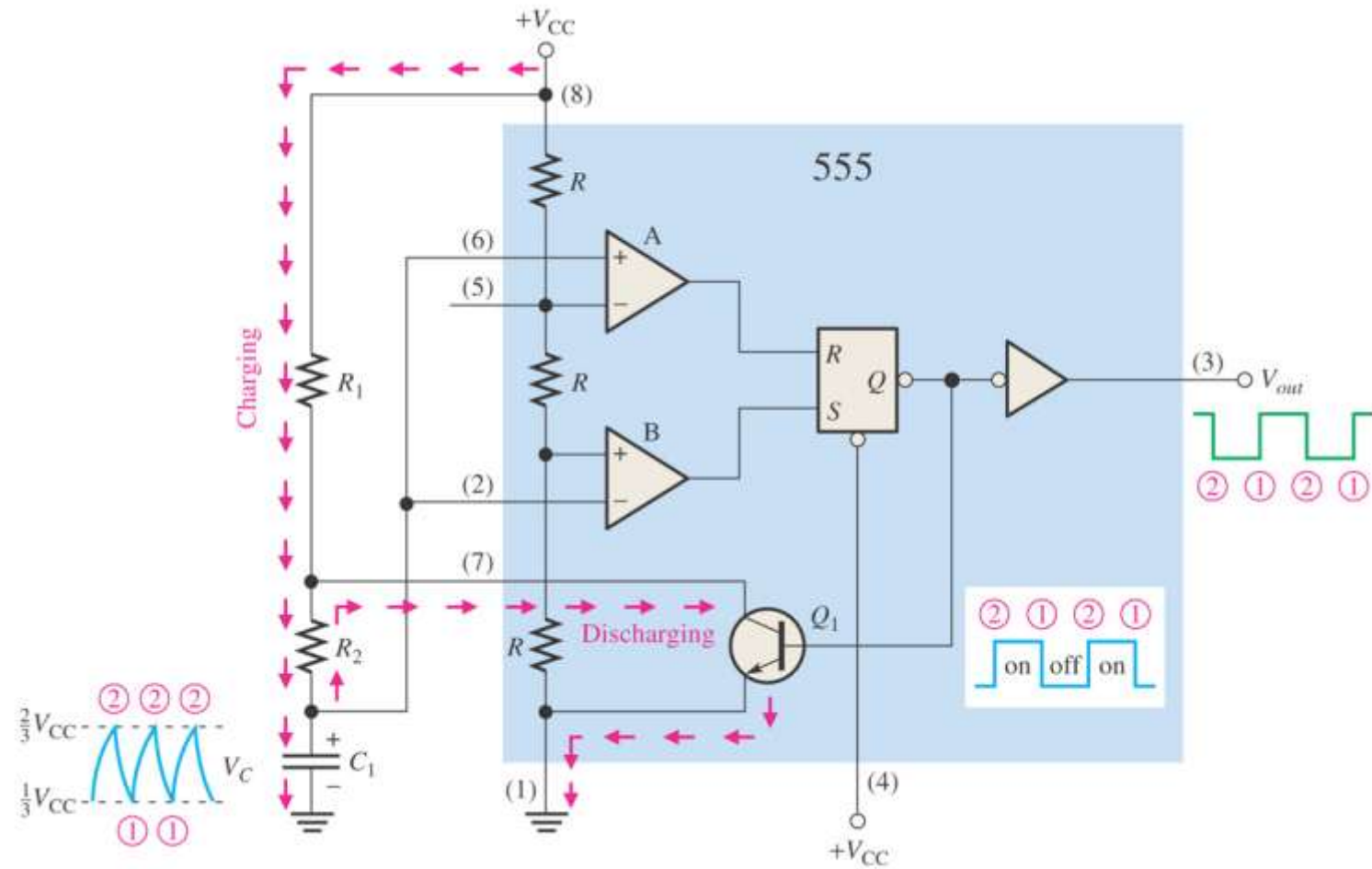
(c) At end of charging interval

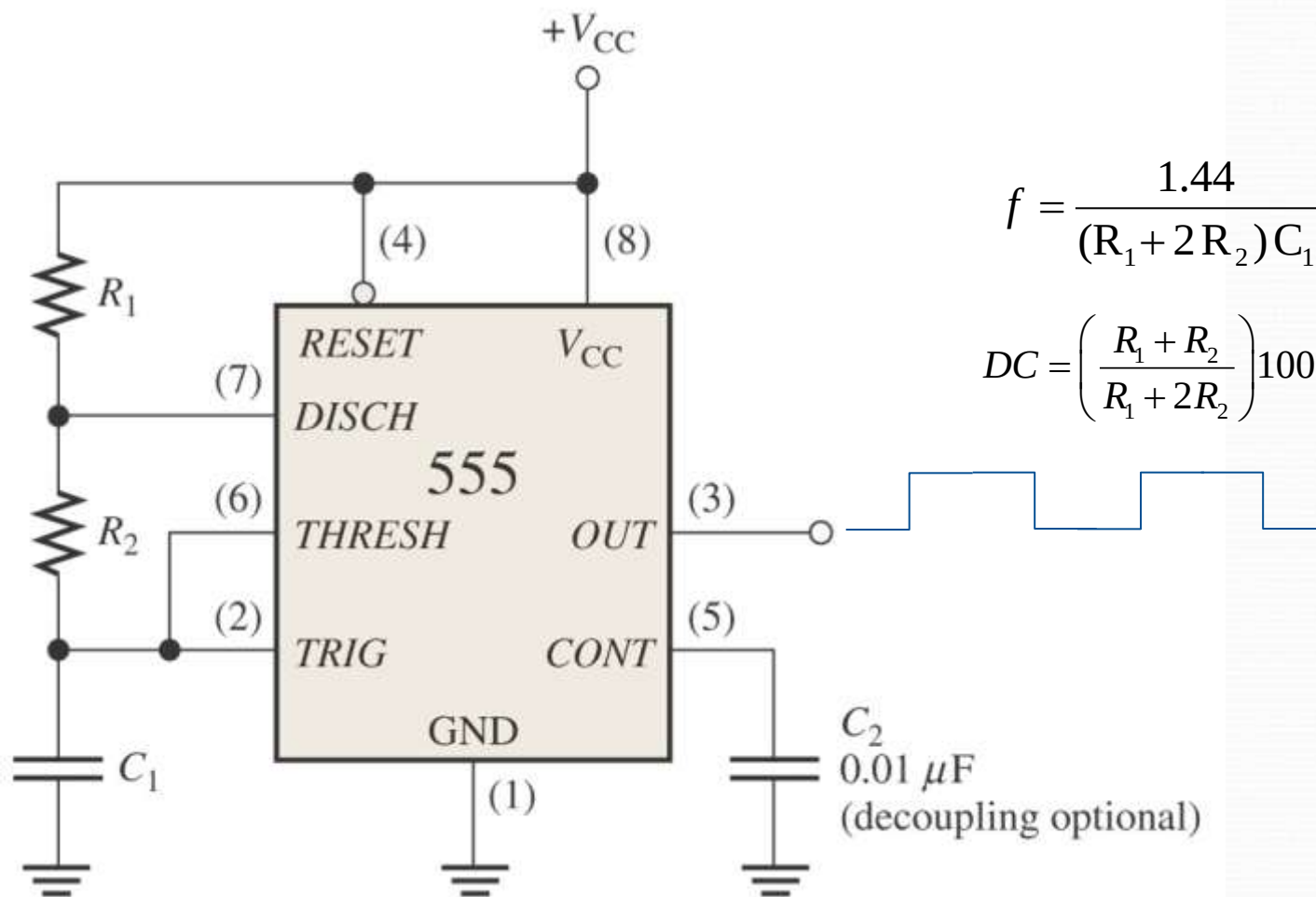


**Figure 7–56** The 555 timer connected as an astable multivibrator (oscillator).



**Figure 7–57** Operation of the 555 timer in the astable mode.





$$f = \frac{1.44}{(R_1 + 2R_2)C_1}$$

$$DC = \left( \frac{R_1 + R_2}{R_1 + 2R_2} \right) 100\%$$

# Summary

- Sequential circuits
- Some basic sequential circuits
  - Latches
  - Edge-triggered Flip-Flops
  - Flip-Flop One-shots
  - The 555 timer

# HW

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